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KNOTLESS SUTURE ASSEMBLY

Abstract

A knotless suture assembly may include a repair suture which may have a first and a second end. The first end may be coupled to a first needle. The knotless suture assembly may also include a shuttle suture. The shuttle suture may have a first end coupled to the first needle and a second end coupled to a second needle. The shuttle suture may include a shuttling feature, and the first end of the repair suture may be configured to engage the shuttling feature during a shuttling process. The knotless suture assembly may include a locking feature. The repair suture may be configured to engage the locking feature during the shuttling process. The locking feature may be configured to retain the repair suture under tension in response to the shuttling process.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119 (e) and the benefit of U.S. Provisional Application No. 63/563,037 entitled KNOTLESS VALVE SUTURE, filed on Mar. 8, 2024, U.S. Provisional Application No. 63/682,911 entitled KNOTLESS VALVE SUTURE, filed on Aug. 14, 2024, and U.S. Provisional Application No. 63/753,635 entitled KNOTLESS SUTURE ASSEMBLY WITH PREFORMED SPLICED SEGMENTS, filed on Feb. 4, 2025, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure generally relates to a suture assembly and, more particularly, to a knotless suture assembly including a repair suture and shuttle suture for cardiovascular valve procedures. During cardiovascular procedures, such as valve repairs or valve replacements, sutures can be used to attach a valve to tissue. The sutures are often knotted to hold the suture under tension. In other aspects, a crimp can be placed over the suture and crimped or deformed to hold the suture under tension. The sutures can form loops or stitches through a shuttling process. In various implementations, the disclosure provides for devices and methods that may assist in the operation of suture assemblies by improving the locking structure and/or improving how the suture is carried through tissue.

SUMMARY

[0003] In various aspects, knotless suture assemblies may be utilized to connect valves to tissue, connect multiple tissue bodies, and close cuts or openings in tissue. Knotless suture assemblies may include a repair suture, a shuttle suture, and one or more needles coupled to the repair suture and/or the shuttle suture. The needle(s) may be used to carry the sutures through tissue and/or a replacement valve concurrently. The sutures may be coupled with a pledget, which can be positioned against an underside of the tissue. The shuttle suture may be used for carrying the repair suture through a locking feature for providing a knotless stitch configuration.

[0004] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget. A repair suture may have a proximal end fixedly coupled to the pledget and a distal end, which may be coupled to a first needle. A shuttle suture may extend through the pledget. The shuttle suture may be coupled with the first needle and a second needle. The shuttle suture may include a shuttling feature proximate to the second needle. A locking feature may be coupled to the pledget. The shuttling feature may be a shuttle link, a receiving area of the shuttle suture, a shuttle loop, and/or an area formed by a folded shuttle suture. The locking feature may be configured to retain the repair suture under tension when the repair suture is threaded through the locking feature in response to a shuttling process. The locking feature may be a spliced segment of the repair suture and/or a grommet. In certain aspects, the pledget may define a ring shape and multiple suture pairs, which each may include a repair suture and a shuttle suture, may be coupled to the pledget with the ring shape.

[0005] In certain aspects, the knotless suture assembly may include a first end of the shuttle suture may form the shuttling feature. The shuttle suture may be spliced into the repair suture at a first

spliced segment proximate a first end of the repair suture and at a second spliced segment proximate to a second end of the repair suture. A stuffing suture may be spliced into the repair suture between the first and second spliced segments between the repair suture and the shuttle suture. The first and second spliced segments may be spaced from first and second ends of the shuttle suture and/or may include the first and second ends of the shuttle suture. One or both ends of the suture assembly may be coupled with a respective needle for insertion through tissue. The shuttle suture may be configured to guide the repair suture through the first spliced segment which can, consequently, bunch the stuffing suture to lock the repair suture in a stitch configuration.
[0006] In additional aspects, a method of manufacturing the knotless suture assembly including the spliced segments may include cutting suture material to a predefined length. A repair suture may be provided or formed. A shuttle suture may be provided and spliced into the repair suture. The shuttle suture may be spliced into the repair suture at one or more locations, with at least one spliced segment forming a locking feature for retaining the repair suture under tension in response to a shuttling process. Often, a lead end of the shuttle suture may be spliced into the repair suture to form at least one combined or spliced end of the suture assembly for inserting the suture assembly through tissue.
[0007] In various aspects, the knotless suture assembly may be utilized in various cardiovascular and orthopedic procedures. For example, the knotless suture assembly may be utilized in a method of replacing a cardiovascular valve. Additionally or alternatively, the knotless suture assembly may be used in a method of compressing tissue bodies to reduce a gap or opening therebetween.
[0008] These and other features, objects, and advantages of the present disclosure will become apparent upon reading the following description thereof together with reference to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is representative of a cardiovascular valve replacement with sutures to bring a replacement valve to engage an annulus, according to the present disclosure;
[0010] FIG. **2** is representative of an aortic annulus with suture assemblies arranged around the annulus, according to the present disclosure;
[0011] FIG. **3** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a pledget under the tissue, a repair suture, a shuttle suture, and two needles, according to the present disclosure;
[0012] FIG. **4** is a schematic diagram of a suture assembly with a repair suture being threaded through a shuttle link of a shuttle suture, according to the present disclosure;
[0013] FIG. **5** is a schematic diagram of a suture assembly with a repair suture threaded through a shuttle link of a shuttle suture, illustrating a pulling force being applied to the shuttle suture to guide the repair suture, according to the present disclosure;
[0014] FIG. **6** is a schematic diagram of a suture assembly with a repair suture threaded through a spliced section of the repair suture to lock the repair suture, according to the present disclosure;
[0015] FIG. **7** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the folded pledget is illustrated in an expanded state, according to the present disclosure;
[0016] FIG. **8** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the folded pledget is illustrated in a compressed state, according to the present disclosure;

[0017] FIG. **9** is a schematic diagram of a suture assembly including an annular pledget and multiple suture pairs with a repair suture and a shuttle suture, according to the present disclosure;

[0018] FIG. **10** is a schematic diagram of a suture assembly with a repair suture threaded through a shuttle link of a shuttle suture, illustrating a pulling force being applied to the shuttle suture to guide the repair suture through a grommet for locking the repair suture, according to the present disclosure;

[0019] FIG. **11** is a flow diagram of a method of using a knotless suture assembly with a shuttle link in a valve replacement procedure, according to the present disclosure;

[0020] FIG. **12** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the shuttle suture includes a shuttle receiving area;

[0021] FIG. **13** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the repair suture is inserted through the shuttle suture at a shuttle receiving area;

[0022] FIG. **14** is a flow diagram of a method of using a knotless suture assembly with a shuttle receiving area in a valve replacement procedure;

[0023] FIG. **15** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the shuttle suture is folded to form a shuttle loop, according to the present disclosure;

[0024] FIG. **16** is a schematic diagram of a suture assembly extending through tissue and a replacement valve, where the suture assembly includes a folded pledget under the tissue, a repair suture, a shuttle suture, and two needles, where the repair suture is inserted through a shuttle loop of the shuttle suture, according to the present disclosure;

[0025] FIG. **17** is a flow diagram of a method of using a knotless suture assembly with a shuttle loop in a valve replacement procedure, according to the present disclosure;

[0026] FIG. **18** is a schematic diagram of a suture assembly with needles coupled to ends of a repair suture and ends and a mid-region of a shuttle suture spliced into the repair suture, according to the present disclosure;

[0027] FIG. **19** is a schematic diagram of the suture assembly of FIG. **18** with the needles removed and the ends of the repair suture and the shuttle suture separated, according to the present disclosure;

[0028] FIG. **20** is a schematic diagram of a suture assembly with needles coupled to ends of a repair suture, ends and a mid-region of a shuttle suture spliced into the repair suture, and a pledget, according to the present disclosure;

[0029] FIG. **21** is a schematic diagram of a repair suture for a suture assembly, according to the present disclosure;

[0030] FIG. **22** is a schematic diagram of a repair suture and a shuttle suture forming a suture assembly, according to the present disclosure;

[0031] FIG. **23** is a schematic diagram of a suture assembly with a shuttle suture and a stuffing suture spliced into a repair suture, according to the present disclosure;

[0032] FIG. **24** is a method of manufacturing a suture assembly, according to the present disclosure;

[0033] FIG. **25** is a schematic diagram of a suture assembly being inserted through a tissue body, according to the present disclosure;

[0034] FIG. **26** is a schematic diagram of a suture assembly being inserted through adjacent tissue bodies, according to the present disclosure;

[0035] FIG. **27** is a schematic diagram of a suture assembly inserted through adjacent tissue bodies with a spliced region between a repair suture and a shuttle suture arranged below the tissue bodies, according to the present disclosure;

[0036] FIG. **28** is a schematic diagram of a suture assembly inserted through adjacent tissue bodies with a repair suture engaging a shuttle loop of a shuttle suture, according to the present disclosure;

[0037] FIG. **29** is a schematic diagram of a suture assembly with a repair suture in a stitch configuration from a shuttling process, according to the present disclosure;

[0038] FIG. **30** is a schematic diagram of a repair suture providing compression between adjacent tissue bodies, according to the present disclosure;

[0039] FIG. **31** is a schematic diagram of a repair suture tensioned to compress adjacent tissue bodies together, according to the present disclosure;

[0040] FIG. **32** is a schematic diagram of a suture assembly being inserted through a tissue body, according to the present disclosure;

[0041] FIG. **33** is a schematic diagram of a suture assembly being inserted through overlapping tissue bodies, according to the present disclosure;

[0042] FIG. **34** is a schematic diagram of a suture assembly inserted through overlapping tissue bodies with a spliced region between a repair suture and a shuttle suture arranged below an inner layer of the overlapping tissue bodies, according to the present disclosure;

[0043] FIG. **35** is a schematic diagram of a suture assembly inserted through overlapping tissue bodies with a repair suture engaging a shuttle loop of a shuttle suture, according to the present disclosure;

[0044] FIG. **36** is a schematic diagram of a suture assembly with a repair suture in a stitch configuration from a shuttling process, according to the present disclosure;

[0045] FIG. **37** is a schematic diagram of a repair suture tensioned to compress overlapping tissue bodies together, according to the present disclosure; and

[0046] FIG. **38** is a method of compressing tissue bodies together, according to the present disclosure.

DETAILED DESCRIPTION

[0047] In the following description, reference is made to the accompanying drawings, which show specific implementations that may be practiced. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. It is to be understood that other implementations may be utilized and structural and functional changes may be made without departing from the scope of this disclosure.

[0048] With reference to FIGS. **1-38**, reference numeral **10** generally designates a knotless suture assembly, exemplary configurations of which are referred to herein as knotless suture assemblies **10**A-**10**E that can provide configurations for different shuttling approaches and/or approaches for insertion through tissue T and/or implant. The knotless suture assembly **10** may include a pledget **12**, which may have various configurations, examples of which are referred to herein as pledgets **12**A-**12**C. The suture assembly **10** may further include a repair suture **14**. The repair suture **14** may include a first end **16** and a second end **18**. The repair suture **14** may be coupled with at least a first needle **20**. A shuttle suture **22** may include a first end **24** and a second end **26** and may extend through the pledget **12**. The shuttle suture **22** may be coupled with at least the first needle **20** or both the first needle **20** and a second needle **28**.

[0049] In certain aspects, the shuttle suture **22** and the repair suture **14** may each be connected to the first needle **20** at a common end of the suture assembly **10**. As illustrated in various examples, the shuttle suture **22** may also form or include a loop, opening, link, or other shuttling feature(s) **30**, which may be proximate to the second needle **28**. Examples of the shuttling feature **30** may be referred to herein as a shuttle link **30**A (see FIGS. **3-8**), a shuttle receiving area **30**B (see FIGS. **12** and **13**), a folded shuttle loop **30**C (see FIGS. **15** and **16**), and an end shuttle loop **30**D (see FIGS. **18**, **19**, **21**, and **22**). Additionally, the knotless suture assembly **10** may include a locking feature **32**. The locking feature **32** may be within or coupled to the pledget **12** and/or part of the repair suture **14** to retain the repair suture **14** under tension.

[0050] The suture assembly **10** may generally configured for use with connecting devices or implants to muscle/soft tissue T, as well as for connecting tissue T bodies together. In a non-limiting example, the suture assembly **10** may be utilized for heart valve replacements or heart valve repairs, where the implant may be a replacement valve V. In such examples, the suture assembly **10** may be utilized for a mitral valve replacement or an aortic valve replacement. Using the aortic valve replacement as an example, an annulus A extends around leaflets of the aortic valve. In the valve replacement, the leaflets are removed from a patient. The suture assembly **10** may be configured to draw the replacement valve V into the patient to implant and replace the valve that was removed from the patient. Depending on the type of valve replacement, the valve replacement procedure may access the aortic valve via the aorta or the mitral valve via the septum.

[0051] The replacement valve V may be a mechanical valve or may be a tissue-based valve, which can often be a xenograft-type of tissue. The type of valve V may be chosen based on various factors, including patient age, patient diagnosis, prior health complications, medications, etc. The suture assembly **10** may be used with any type of replacement valve V without departing from the teachings herein. The suture assembly **10** may assist with defining a fluid-tight or blood-tight seal between the tissue T and the replacement valve V. Moreover, the suture assembly **10** may be re-tensioned, as described herein, during the valve replacement procedure to maximize alignment of the replacement valve V.

[0052] Referring to FIGS. **1** and **2**, during the valve replacement procedure, the replacement valve V can be brought to engage and compress against the tissue T. The replacement valve V may generally be sized to correspond with a size of the annulus A or a tube that the replacement valve V is being sewn into. For example, if the opening for the tube has a 27 mm diameter, a 27 mm replacement valve V may be used to reduce or prevent any gaps outside of the replacement valve V.

[0053] Multiple suture assemblies **10**, including any one or more of the exemplary configurations disclosed herein, may be used to secure the replacement valve V to the tissue T. For example, between about 12 and about 17 suture pairs **40**, each generally including one repair suture **14** and one shuttle suture **22**, may be utilized. The suture pairs **40** can be spaced around the annulus A to secure the replacement valve V to the tissue T. The knotless aspect of the suture assembly **10** may be advantageous for more efficiently securing the replacement valve V to the tissue T while reducing components arranged on the replacement valve V, where conventional knots and crimps can be positioned.

[0054] Referring to FIG. **3**, the suture assembly **10** may include the repair suture **14** and the shuttle suture **22**. The repair suture **14** may be used to attach the replacement valve V to the tissue T, while the shuttle suture **22** may be utilized to shuttle or guide the repair suture **14** to form a stitch upon the application of a pulling force or tension. The configuration of the shuttle suture **22** and/or the arrangement of the shuttle suture **22** relative to the pledget **12** may provide different shuttling approaches. The suture assembly **10** may also include the pledget **12**, and each of the repair suture **14** and the shuttle suture **22** may be operably coupled with or engaged with the pledget **12**.

[0055] The first end **16** of the repair suture **14** may be coupled with the first needle **20**. In this configuration, the first end **16** may also be referred to as a distal end, a tail end, or a tail. The distal end **16** may be spaced from the pledget **12** to provide a length for the repair suture **14** to form the stitch, be tensioned, and be re-tensioned. In various aspects, the repair suture **14** may have a greater thickness or width at the proximal end **18** and a tapered width at the distal end **16**. The proximal end **18** may include stuffing **44** or stuffing suture **44** (see FIG. **5**), which may be additional suture material stuffed within the repair suture **14** to increase the width or thickness to assist with locking the repair suture **14**. The distal end **16** may be tapered through the manufacturing process. For example, in aspects where the repair suture **14** is woven together, fewer strands of material may be used to form the distal end **16**. The tapered distal end **16** may assist with the distal end **16** being carried by the shuttle suture **22** during the shuttling process as described later herein with reference to FIGS. **4-6**.

[0056] In various aspects, the repair suture **14** may include the second end **18**, which may be referred to as a proximal end, a base, or a base end. The proximal end **18** may be fixed to the pledget **12** via a base knot **42**. The base knot **42** may be positioned proximate to a first side of the pledget **12**, and the repair suture **14** may extend along the pledget **12** to exit the pledget **12** proximate to an opposing second side.

[0057] Referring still to FIG. **3**, for the illustrated knotless suture assembly **10**A, the shuttle suture **22** may include the first end **24**, which may be referred to as a tensioning end, and the second end **26**, which may be referred to as a guiding or shuttling end. The ends **24**, **26** of the shuttle suture **22** may both be spaced from the pledget **12**, while a central or middle section of the shuttle suture **22** may extend through the pledget **12**. The first end **24** of the shuttle suture **22** may be coupled with the first needle **20**. In this way, both the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** may be coupled to the first needle **20**. The first end **24** of the shuttle suture **22** and the distal end **16** of the repair suture **14** may both be coupled to the first needle **20** such that there are two tail ends coupled with the needle **20**. Alternatively, the first end **24** of the shuttle suture **22** may be spliced into the repair suture **14** for a single tail (e.g., the distal end **16** or the combined ends **16**, **24**) to be coupled to the needle **20**, similar to as described later herein with respect to FIGS. **18-20** and FIGS. **22-24**.

[0058] Additionally, the second end **26** of the shuttle suture **22** may be coupled with the second needle **28**, such that both ends **24**, **26** of the shuttle suture **22** may be coupled with needles **20**, **28**, respectively. The second end **26** of the shuttle suture **22** may define or include the shuttle link **30**A, which may be an opening or loop through which the repair suture **14** can be inserted and wrapped to pass through the locking feature **32** as later described in reference to FIGS. **4-6**. In non-limiting examples, the shuttle link **30**A may be formed by a bifurcated segment of the shuttle suture **22** at the second end **26**. In such examples, during the manufacturing process, the shuttle suture **22** may be woven, braised, interlaced, or otherwise formed as a single limb, then formed as two limbs to form the bifurcated segment, and then again formed as the single limb to form the shuttle link **30**A. This process may be advantageous for retaining a smaller width of the shuttle suture **22** at the shuttle link **30**A. With the bifurcated segment forming the shuttle link **30**A, the width of the bifurcated segment may be substantially similar to the remainder of the shuttle suture **22**. Additionally, the shuttle suture **22** may have a single tail adjacent to the shuttle link **30**A for engaging the needle **28**.

[0059] In additional or alternative examples, the shuttle link **30**A may be formed as a loop (see, e.g., FIGS. **19**, **22**, and **23**). In such examples, the second end **26** of the shuttle suture **22** may be folded to engage itself to form the loop. In this configuration, the width of the shuttle suture **22** at the loop may be greater than the width of the remainder of the shuttle suture **22**. Alternatively, the shuttle suture **22** may be manufactured to taper at the second end **26** to reduce the size of the loop. For example, the tapered end **26** forming the loop may have a width or thickness substantially equal to the remainder of the shuttle suture **22**.

[0060] Referring still to FIG. **3**, one or both of the repair suture **14** and the shuttle suture **22** may have cut markers or indicators **50-54** proximate to the needles **20**, **28**. The repair suture **14** may have the cut indicator **50** at the distal end **16** proximate to the first needle **20**. The shuttle suture **22** may have cut indicators **52**, **54** at each of the first end **24** and the second end **26**, respectively. At the second end **26**, the cut indicator **54** may be between the shuttle link **30**A and the second needle **28**.

[0061] The cut indicators **50-54** may provide a location for separating the needles **20**, **28** from the sutures **14**, **22** after the suture assembly **10** is engaged with the tissue T and/or the implant/replacement valve V. The cut indicators **50-54** may generally leave a similar length of the repair suture **14** and two lengths of the shuttle suture **22** extending from the replacement valve V to be used to form the stitch. Upon removal of the needles **20**, **28**, the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** may be separated from one another. Further, depending on the configuration of the shuttle link **30**A, upon removal of the second needle **28**, the second end **26** of the shuttle suture **22** may end in a tail section extending from the bifurcated segment or may end in the loop.

[0062] Referring still to FIG. **3**, the pledget **12** may generally be a mesh-like or braided component that can assist with distributing pressure from the sutures **14**, **22** along a greater surface area of the tissue T. The pledget **12** may form a buttress for the suture assembly **10** against the tissue T. In this configuration, the repair suture **14** may be fixed to the pledget **12** and may extend at least partially through the pledget **12**. The shuttle suture **22** may extend through the pledget **12** and can be removed from the pledget **12** during the shuttling process. The braided or mesh configuration of the pledget **12** may allow the sutures **14**, **22** to be passed through the pledget **12** to form the stitch.

[0063] Referring now to FIGS. **3** and **4**, for example, when used for a replacement valve procedure, the pledget **12** may be arranged along an inner surface of the tissue T or under the annulus A. The suture assembly **10**A may include two needles **20**, **28** provided to pass the ends **16**, **18**, **24**, **26** of the sutures **14**, **22** through the tissue T from a first side to an opposing second side and through the replacement valve V. The first needle **20** can be inserted through the tissue T from the underside of the annulus A and then through the replacement valve V, which may bring the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** through the tissue T and the replacement valve V. Similarly, the second needle **28** can be inserted through the tissue T from the underside of the annulus A and then through the replacement valve V, which may bring the second end **26** of the shuttle suture **22** through the tissue T and the replacement valve V. The sutures **14**, **22** extending through the tissue T may be spaced by approximately the length of the pledget **12**.

[0064] The use of two needles **20**, **28** may allow the suture assembly **10** to be packaged in a preassembled configuration ready for application in a sterile setting. The two needles **20**, **28** and corresponding engagement to the sutures **14**, **22** may allow the pledget **12** to remain under the annulus A to distribute pressure caused by the sutures **14**, **22**. Further, the use of two needles **20**, **28** can assist with strengthening an engagement between the tissue T and the replacement valve V to provide a better seal and compression of the replacement valve V into the tissue T. For example, with the aortic valve replacement, there is generally a high gradient of blood flow that moves through the aortic valve into the body of the patient to be able to pump blood through the extremities of the patient. The two pass-throughs of the suture assembly **10** through the tissue T and the replacement valve V may assist in providing a blood-tight seal between the tissue T and the replacement valve V with greater compression of the replacement valve V against the tissue T.

[0065] Referring to FIGS. **4-6**, once the needles **20**, **28** have been removed, the sutures **14**, **22** may be utilized to form the stitch typically via the shuttling process. The distal end **16** of the repair suture **14** may be threaded through the shuttle link **30**A. The repair suture **14** may include one or more fold indicators **56**, which may be visually differentiated from the cut indicators **50-54** to guide a user in where to cut and where to fold the repair suture **14**. The distal end **16** of the repair suture **14** may be inserted through the shuttle link **30**A and folded at the fold indicator **56**.

[0066] The repair suture **14** may generally be folded back to abut and extend along itself to form the folded distal end **16** around the shuttle link **30**A. This folded arrangement around the shuttle link **30**A may allow the repair suture **14** to remain engaged with the shuttle link **30**A as the pulling force is applied to the shuttle suture **22**. The distal end **16** of the repair suture **14** being narrower or tapered may be advantageous for forming a smaller folded distal end **16**. The smaller folded distal end **16** may be advantageous for guiding the folded distal end **16** through the pledget **12** and/or the locking feature **32** while reducing resistance.

[0067] Still referring to FIGS. **4-6**, the pulling force can be applied to the first tensioning end **24** of the shuttle suture **22**, which may draw the shuttle link **30**A and the engaged folded distal end **16** of the repair suture **14** toward and through the replacement valve V and the tissue T. As the proximal end **18** of the repair suture **14** can be coupled to the pledget **12** at the base knot **42**, the proximal end **18** may remain in a generally fixed location as the repair suture **14** is looped over the replacement valve V and then guided through the tissue T and the replacement valve V. At this stage, the repair suture **14** may extend through the tissue T and the replacement valve V at two different locations and can extend across the replacement valve V between the two locations. Though discussed primarily in reference to the replacement valve V, it shall be understood that knotless suture assembly **10** may be applied in a variety of applications and procedures.

[0068] As in the configuration illustrated in FIG. **5**, the suture assembly **10**A may include the locking feature **32**, which may be configured as a spliced section **70** of the repair suture **14** proximate to the base knot **42**. Generally, the spliced section **70** may be disposed within the pledget **12**. At the spliced section **70**, the shuttle suture **22** may be woven, interlaced, or otherwise extended through the repair suture **14**. The repair suture **14** may include the stuffing **44** at or proximate to the spliced section **70**. In this configuration, the tension applied to the repair suture through the increased width of the stuffing **44** or padding may cause the stuffing **44** to compress or wrinkle to assist with binding or locking the repair suture **14**. In the instant example, the spliced section **70** may be between about five millimeters and about six millimeters in length along the repair suture **14**. However, the proportions of the spliced section **70** and various features of the suture assembly **10** may vary depending on the application.

[0069] With the tension or pulling force being applied to the first end **24** of the shuttle suture **22**, the shuttle suture **22** may be drawn through the spliced section **70**. As the shuttle suture **22** is drawn through the spliced section **70**, the folded distal end **16** of the repair suture **14** may also be drawn through the spliced section **70**. Accordingly, the repair suture **14** may be pulled through itself at the spliced section **70**. The tapered distal end **16** may be advantageous for pulling the repair suture **14** through itself with less resistance. The pulling force may cause the shuttle suture **22** to be pulled entirely through the spliced section **70** and may pull the repair suture **14** through the spliced section **70** until the repair suture **14** is tightened or tensioned.

[0070] In certain aspects, the padding or stuffing **44** may be added at the base of the repair suture **14** proximate to the base knot **42** and the spliced section **70** to assist with locking the repair suture **14** as the repair suture **14** extends through itself at the spliced section **70**. As the repair suture **14** is pulled through itself, the thicker part of the repair suture **14** can bunch or catch to create a locking trap. This locking feature **32** and the resistance caused by the thicker, stuffed portion of the repair suture **14** may lock the repair suture **14** as the repair suture **14** is tensioned.

[0071] As illustrated in FIGS. **5** and **6**, the pulling force can continue to be applied to the shuttle suture **22** to draw the shuttle suture **22** and the folded distal end **16** of the repair suture **14** through the pledget **12**. As shown in FIG. **5**, the shuttle suture **22** and the folded distal end **16** of the repair suture **14** can be drawn across the pledget **12**, through the tissue T, and through the replacement valve V by the pulling force on the first end **24** of the shuttle suture **22**. As shown in FIG. **6**, the shuttle suture **22** may be pulled fully from the tissue T and the replacement valve V. The withdrawal of the shuttle suture **22** may further cause the shuttle link **30**A to draw the distal end **16** of the repair suture **14** along the same path, such that the distal end **16** is exposed from the tissue. Accordingly, the entire path of the closure loop formed by the repair suture **14** may be as follows: the repair suture **14** may extend from the base knot **42**, through the pledget **12** under the annulus A, through the tissue T and the replacement valve V, over the replacement valve V, back through the replacement valve V and the tissue T, through the spliced section **70**, and again through the tissue T and the replacement valve V to form the stitch. The resulting exposed distal end **16** of the repair suture **14** may be pulled to tension the suture **14**, and the locking feature **32** may automatically lock the suture **14** as the repair suture **14** is tensioned.

[0072] The shuttle suture **22** and the repair suture **14** may be pulled through the replacement valve V and the tissue T in the locations where the sutures **14**, **22** are initially inserted with the needles **20**, **28**. This may be advantageous for reducing the length of the repair suture **14** is utilized. Additionally, this may be advantageous for fully removing the shuttle suture **22** from the patient during the replacement procedure.

[0073] The pledget **12**, as illustrated in FIGS. **4-6**, may generally be rectangular or square in profile and may have a thin, generally flat body, referred to as the flat pledget **12**A. In such configurations, the pledget **12**A may be formed of a woven, mesh, or braided structure through which the repair suture **14** and the shuttle suture **22** extend and travel. The pledget **12** may be constructed of polytetrafluoroethylene, such as Teflon®, or may be constructed of a fiber material, such as polyester.

[0074] As illustrated in FIGS. **7** and **8**, the pledget **12** may be a braided tubular component that can be folded, which may be referred to herein as a folded pledget **12**B. The folded pledget **12**B may be a braided tube that is folded to form more than one layer **74**, **76**. As in the illustrated configuration, the folded pledget **12**B may include two layers **74**, **76** with openings into the tubular structure being oriented in a same direction (illustrated as a right direction). Accordingly, the folded pledget **12**B may form the first layer **74** which can engage the tissue T, and the second layer **76** which can engage the first layer **74**. The folded pledget **12**B may be constructed of polytetrafluoroethylene or polyester. Additionally, the folded pledget **12**B may provide additional density and depth for forming the locking feature **32**.

[0075] In configurations of the suture assembly **10** with the folded pledget **12**B, the proximal end **18** of the repair suture **14** may be coupled to the first layer **74** via the base knot **42** proximate to a first end of the folded pledget **12**B. The first end of the folded pledget **12**B may be the end forming the openings into the tubular structure, as illustrated, or, alternatively, an end connecting the two layers **74**, **76**. The repair suture **14** may be configured to extend from the base knot **42**, through the second layer **76** of the folded pledget **12**B, along the second layer **76**, through the second layer **76**, through the first layer **74**, and through the tissue T and the replacement valve V to the first needle **20**.

[0076] The shuttle suture **22** may extend from the first needle **20**, through the replacement valve V and the tissue T, through the first layer **74** proximate to the second end of the pledget **12**B, through the second layer **76**, along the second layer **76**, through the second layer **76** proximate to the first end **24**, through the first layer **74**, and through the tissue T and the replacement valve V to the second needle **28**. In certain aspects, the repair suture **14** and/or the shuttle suture **22** may not extend fully out of the second layer **76** and back into the second layer **76**. In such examples, the repair suture **14** and/or the shuttle suture **22** may extend through the second layer **76** of the folded pledget **12**B between the first and second ends of the folded pledget **12**B, such as within a lumen defined by the folded pledget **12**B.

[0077] In response to the needles **20**, **28** being inserted through the tissue T and the replacement valve V, the sutures **14**, **22** may be pulled to compress the folded pledget **12**B for the second layer **76** to abut and press against the first layer **74**, as illustrated in FIG. **8**, in the configuration with the folded pledget **12**B, the sutures **14**, **22** may extend through a middle of the folded pledget **12**B rather than the openings into the tubular structure. The openings are oriented in a first direction and the sutures **14**, **22** may extend through the folded pledget **12**B in a second direction, which may be normal to the first direction.

[0078] The suture assembly **10**A illustrated in FIGS. **7** and **8** may be used in a substantially similar manner as described with respect to FIGS. **3-6**. The needles **20**, **28** may be removed from the sutures **14**, **22**, and the distal end **16** of the repair suture **14** may be inserted through the shuttle link **30**A and folded to form the folded distal end **16**. The pulling force may be applied to the second end **26** of the shuttle suture **22** to draw the shuttle link **30**A and the folded distal end **16** through the tissue T and replacement valve V, through the pledget **12**, and again through the tissue T and the replacement valve V to form the tensioned stitch.

[0079] Referring again to FIGS. **3-8**, the suture assembly **10**A may include the spliced section **70** in the repair suture **14**, which may be in the pledget **12** or proximate to the pledget **12**, such as along the second layer **76**. The shuttle suture **22** may guide the repair suture **14** through itself at the spliced section **70** to lock the tensioned stitch as described herein. The locking feature **32** may correspond to the thickened portion of the spliced section **70** of the repair suture **14** that forms the locking trap as the repair suture **14** is drawn through itself. This configuration may be advantageous for reducing components in the suture assembly **10** and reducing components that remain in the patient. Further, this configuration may reduce or eliminate metal components, such as conventional crimps, from the suture assembly **10**. The reduction or elimination of metal components may be advantageous for any subsequent X-ray or magnetic resonance imaging (MRI) where metal can affect the imaging process.

[0080] Referring to FIG. **9**, in additional or alternative examples, the suture assembly **10** may include a ring-shaped or annular pledget **12**C. The annular pledget **12**C may be configured to be positioned around the annulus A. The suture assembly **10** may include suture pairs **40** to have multiple repair sutures **14** each with an associated shuttle suture **22**. Accordingly, a single annular pledget **12**C may be used with multiple repair sutures **14** and shuttle sutures **22** arranged in the suture pairs **40** (e.g., one repair suture **14** and one shuttle suture **22**) along the ring formed by the pledget **12**.

[0081] In this configuration, the annular pledget **12**C may support most or all of the suture pairs **40** for the replacement valve procedure. For example, the annular pledget **12**C may support between 12 and 17 suture pairs **40**, which results in between 24 and 34 sutures **14**, **22** and needles **20**, **28**. This configuration may be advantageous for maximizing efficiency when adding multiple suture pairs **40** around the annulus A during the valve replacement procedure. Additionally, this configuration may be advantageous for further distributing pressure along the annulus A from the tensioned sutures **14** and providing the compression between the tissue T and the replacement valve V. Each suture pair **40** may operate as described herein. Additionally, the suture assembly **10** may include multiple locking features **32**, which may be in the form of the spliced section **70**, a grommet **80** (see FIG. **10**), or a combination thereof. The pledget **12** may also form different shapes or lengths that support different numbers of suture pairs **40** without departing from the teachings herein.

[0082] Referring to FIG. **10**, in additional or alternative configurations, the locking feature **32** may be configured as a grommet **80** disposed in or coupled to the pledget **12**. In certain aspects, the grommet **80** may be disposed in the mesh or braided material that forms the pledget **12**. The grommet **80** may include inner projections or barbs **82** that may allow for the sutures **14**, **22** to be passed in one direction and not the opposing direction. The inner barbs **82** may be angled to allow movement through the grommet **80** to tension the repair suture **14** and to reduce or prevent movement that would loosen the repair suture **14**, locking the tensioned repair suture **14**.

[0083] Additionally, the grommet **80** may include outer projections or barbs **84** that engage the braided structure of the pledget **12**. This engagement may retain the grommet **80** in a predefined location relative to the pledget **12**, including when the pulling force is applied to the sutures **14**, **22** being pulled through the grommet **80**. The fixed location of the grommet **80** generally assists with more fully tensioning the repair suture **14** and locking the repair suture **14** under tension.

[0084] In the configuration with the grommet **80**, at least the shuttle suture **22** may initially extend through the grommet **80**. The repair suture **14** may extend from the base knot **42** through the pledget **12** proximate to the grommet **80** or may extend through the grommet **80**. As the shuttle suture **22** extends through the grommet **80**, when the pulling force is applied to the shuttle suture **22**, the shuttle link **30**A may draw the folded distal end **16** of the repair suture **14** through the grommet **80** to then lock the repair suture **14** under tension.

[0085] While the grommet **80** is illustrated in FIG. **10** with the flat pledget **12**A, the grommet **80** may be used in the tubular folded pledget **12**B and/or the annular pledget **12**C without departing from the teaching herein. In the tubular pledget **12**B configuration, the grommet **80** may be disposed within the tubular structure (e.g., in an interior channel or lumen formed by the tube) or coupled to at least one of the layers **74**, **76** of the folded pledget **12**B.

[0086] Referring to FIG. **11**, and with further reference to FIGS. **1-10**, a method **100** of using the suture assembly **10**A for attaching an implant to tissue, such as in a valve replacement includes providing the suture assembly **10** (**102**) that includes the pledget **12**, at least one repair suture **14** and associated shuttle suture **22**, and the two needles **20**, **28**. As previously described, the leaflets of the aortic valve may be removed from the patient, and the replacement valve V may be provided. The pledget **12** can be arranged on the underside of the annulus A (**104**) where the replacement valve V is to be attached. The first needle **20** can be inserted through the tissue T and the replacement valve V (**106**), carrying the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** through the tissue T and the replacement valve V.

[0087] The second needle **28** may be inserted through the tissue T and the replacement valve V (**108**), carrying the second end **26** of the shuttle suture **22** through the tissue T and the replacement valve V. At this stage, the ends of the sutures **14**, **22** and the needles **20**, **28** may be on an opposing side of the tissue T and the replacement valve V compared to the pledget **12**. Tension may be applied by pulling both needles **20**, **28** to move the pledget **12** to engage the tissue T.

[0088] The sutures **14**, **22** may be cut (**110**) at the cut indicators **50-54** to remove the needles **20**, **28** from the sutures **14**, **22**. The distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** may be separated from one another with the removal of the first needle **20**. The distal end **16** of the repair suture **14** may be inserted or threaded through the shuttle link **30**A (**112**). The distal end **16** may be folded at the fold indicator(s) **56** to form the folded distal end **16** around the shuttle link **30**A.

[0089] The pulling force is applied to the first end **24** of the shuttle suture **22** (**114**), which may draw the folded distal end **16** of the repair suture **14** and the shuttle link **30**A through the replacement valve V and the tissue T. The pulling force may draw the sutures **14**, **22** into or through the pledget **12** and through the locking feature **32** (**116**). In configurations with the spliced section **70**, the shuttle link **30**A and the folded distal end **16** may be drawn through the spliced section **70**. In such configurations, the repair suture **14** may be pulled through itself. The thickened portion of the repair suture **14** proximate to the spliced section **70** may bunch to lock the repair suture **14**. In configurations with the grommet **80**, the shuttle link **30**A and the folded distal end **16** may be drawn through the grommet **80**. The pulling force may continue to be applied to the shuttle suture **22** so the shuttle link **30**A and the folded distal end **16** are drawn through the tissue T and the replacement valve V (**118**) proximate to an opposing side of the pledget **12**. The shuttle suture **22** may be fully removed from the tissue T and the replacement valve V.

[0090] The repair suture **14** may continue to be pulled to form the stitch and tension the repair suture **14** (**120**). Additional suture assemblies **10** and/or suture pairs **40** may follow this process (e.g., steps **102-120** with multiple pledgets **12** or steps **106-120** with the annular pledget **12**C) to attach the replacement valve V to the tissue T. One or more of the repair sutures **14** may be re-tensioned (**122**). An additional pulling force may be applied to the distal ends **18** of the repair sutures **14** to re-tension the repair sutures **14**. The locking feature(s) **32** may automatically lock the repair sutures **14** with each re-tensioning.

[0091] A pusher/cutter device may be used to apply pressure on the replacement valve V while tensioning (**120**) and re-tensioning (**122**) the repair suture **14**. This may be advantageous for reducing displacement of the replacement valve V and reducing slack from the suture assemblies **10**. The repair sutures **14** may be cut (**124**) to be flush with the replacement valve V. The use of the pusher/cutter device may provide a better seal between the tissue T and the replacement valve V, as well as more fully tension the repair sutures **14** to then be cut flush with the replacement valve V. The locking features **32** retain the repair sutures **14** in the stitch configuration under tension. The steps **102-124** of the method **100** may be modified or conducted in different orders, with steps omitted, repeated, performed concurrently, etc. without departing from the teachings herein.

[0092] Referring to FIGS. **12** and **13**, the suture assembly **10**B is illustrated, which may provide an additional shuttling approach compared to the suture assembly **10**A described in reference to FIGS. **1-11**. The suture assembly **10**B may include the pledget **12**, the repair suture **14**, the shuttle suture

22, and the locking feature **32**. The shuttling approach provided by the suture assembly **10**B may be based primarily on the configuration of the shuttle suture **22**. Accordingly, the pledget **12**, the repair suture **14**, and the locking feature **32** may be substantially similar to any of the configurations described with reference to FIGS. **1-11**.

[0093] The pledget **12**, illustrated as the folded pledget **12**B, may be configured to support the repair suture **14** and the shuttle suture **22**. The pledget **12** in the suture assembly **10**B may be any configuration, such as the flat pledget **12**A, the folded pledget **12**B, or the annular pledget **12**C without departing from the teachings herein. Similar to the configuration described above, the proximal end **18** of the repair suture **14** may be coupled to the pledget **12** by the base knot **42**. The repair suture **14** may have the increased thickness at or proximate to the proximal end **18**, which may be from the stuffing **44**. As previously set forth, the stuffing **44** may be additional suture material stuffed within the repair suture **14** to increase the width or thickness and assist with locking the repair suture **14**. The repair suture **14** may extend through and out of the pledget **12** to the distal end **16**. The distal end **16** of the repair suture **14** may taper in width, which may assist with drawing the repair suture **14** through the locking feature **32** and the pledget **12** during the shuttling process. The distal end **16** may be coupled with the first needle **20** for carrying the distal end **16** through the tissue T and the valve V.

[0094] As in the illustrated configuration of FIGS. **12** and **13**, the locking feature **32** may be configured as the spliced section **70**. At the spliced section **70**, the repair suture **14** generally forms two limbs, and the shuttle suture **22** may be configured to extend between or be woven through the two limbs. This arrangement may allow the shuttle suture **22** to draw the distal end **16** of the repair suture **14** through the spliced section **70** of the repair suture **14**. In response to the repair suture **14** being pulled through itself, the thicker part of the repair suture **14** with the stuffing **44** can bunch or catch to create the locking trap. The spliced section **70** and the bunched stuffing **44** may lock the repair suture **14** in the stitch configuration in response to tension applied to the opposing ends **16**, **18**. In this way, the repair suture **14** may be secured to retain the stitch configuration that couples the replacement valve V to the tissue T without requiring additional hardware or knots. Additionally, the grommet **80** (FIG. **10**) may be similarly implemented to retain the repair suture **14** using the inner barbs **82**.

[0095] The suture assembly **10**B also includes the shuttle suture **22** for shuttling or guiding the repair suture **14**, which may include the first end **24** coupled with the first needle **20** and the second end **26** coupled to the second needle **28**. In this way, both the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** may be coupled to the first needle **20**. Additionally, both ends **24**, **26** of the shuttle suture **22** may be coupled with needles **20**, **28**, respectively, for being carried through the tissue T and the implant/valve V. The first and second ends **24**, **26** may be on opposing sides of the pledget **12** such that the middle section of the shuttle suture **22** extends through the pledget **12** and/or the locking feature **32**.

[0096] Referring still to FIGS. **12** and **13**, the shuttle suture **22** in the suture assembly **10**B may be configured to shuttle or guide the repair suture **14** without including a specific component for receiving the repair suture **14**. The shuttle suture **22** may include the shuttling feature **30**, which may be configured as the shuttle receiving area **30**B. The shuttle receiving area **30**B may be one or more locations or regions on the shuttle suture **22** configured to receive the repair suture **14**. Stated differently, the shuttle receiving area **30**B may be an area or region through which the repair suture **14** is configured to be inserted for the shuttling process. In certain aspects, the shuttle receiving area **30**B may be any location along the shuttle suture **22** between the pledget **12** and the second end **26** (e.g., location(s) on the shuttle suture **22** on an opposing side of the pledget **12** compared to the distal end **16** of the repair suture **14**).

[0097] In additional or alternative examples, the shuttle receiving area **30**B may be a predefined location or region that may maximize the efficiency of the shuttling process. In such examples, the shuttle receiving area **30**B may be proximate to the second end **26** and the second needle **28**, which may maximize the efficiency of the shuttling process and reduce excess length of the repair suture **14** in the pledget **12** in response to the repair suture **14** being tensioned. In the examples with a predefined location or region, the predefined location or region may be marked on the shuttle suture **22** with one or more identifiers **130** or other indicating marks.

[0098] In the configuration illustrated in FIGS. **12** and **13**, the shuttle suture **22** may not have a preformed opening or loop through which the repair suture **14** is configured to be inserted. In this regard, the shuttle suture **22** may be a single limb from the first end **24** coupled with the first needle **20** to the second end **26** coupled with the second needle **28**. This single limb may result in a substantially consistent thickness of the shuttle suture **22** from the first end **24** to the second end **26**. The repair suture **14** may be configured to be inserted through the shuttle suture **22** at the shuttle receiving area **30**B using a suture-through-suture approach. For example, when the shuttle suture **22** is formed by weaving or interlacing multiple yarns or strands of material, the repair suture **14** may be inserted between the different yarns or strands of material that form the shuttle suture **22**. In certain aspects, the repair suture **14** may include one or more insertion indicators **132**, which may be in addition to or in lieu of the fold indicator **56** (FIG. **3**). The insertion indicator **132** may be proximate to the distal end **16** and provide a visual indication of how far to insert the repair suture **14** through the shuttle suture **22** for the shuttling process.

[0099] Referring to FIG. **14**, and with further reference to FIGS. **12** and **13**, the suture assembly **10**B may form the stitch and can be used in a valve replacement procedure or other implant attachment procedure in a substantially similar manner as the suture assembly **10**A, with differences primarily in the shuttling approach. A method **150** of using the suture assembly **10**B for a valve replacement may include providing the suture assembly **10**B (**152**) that includes the pledget **12**, at least one repair suture **14** and associated shuttle suture **22**, and the two needles **20**, **28**. As previously set forth, the leaflets of the aortic valve may be removed from the patient, and the replacement valve V may be provided. The pledget **12** can be arranged on the underside of the annulus A (**154**) where the replacement valve V is attached.

[0100] The first needle **20** can be inserted through the tissue T and the replacement valve V (**156**). With the repair suture **14** and the shuttle suture **22** coupled to the first needle **20**, the first needle **20** may carry the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** through the tissue T and the replacement valve V. Similarly, the second needle **28** may be inserted through the tissue T and the replacement valve V (**158**). The second needle **28** can carry the second end **26** of the shuttle suture **22** through the tissue T and the replacement valve V.

[0101] In response to steps **156** and **158**, the ends **18**, **24**, **26** of the sutures **14**, **22** and the needles **20**, **28** may be on the opposing side of the tissue T and the replacement valve V (e.g., proximate an outer surface of the replacement valve V) compared to the pledget **12** (e.g., abutting an inner surface of the tissue T). Tension may be applied by pulling both needles **20**, **28** to apply tension to the sutures **14**, **22** to move the pledget **12** to engage the tissue T. The sutures **14**, **22** may cause an initial engagement between the tissue T and the valve V, with the ends **18**, **24**, **26** arranged to prepare for the shuttling process to form the seal between the tissue T and the valve V.

[0102] With the ends **18**, **24**, **26** exposed on an outside of the tissue T and the valve V, the shuttle suture **22** may be cut (**160**) at the cut indicators **52**, **54** to remove the needles **20**, **28** from the shuttle suture **22**. However, the distal end **16** of the repair suture **14** may remain coupled with the first needle **20**. Accordingly, the distal end **16** of the repair suture **14** and the first end **24** of the shuttle suture **22** may be separated from one another by cutting the shuttle suture **22**.

[0103] At least the second end **26** of the shuttle suture **22** may be held to move the first needle **20** and, consequently, the distal end **16** of the repair suture **14** relative to the shuttle suture **22**. The first needle **20** may be inserted or threaded through the shuttle suture **22** (**162**), which brings the distal end **16** of the repair suture **14** through the shuttle suture **22**, forming the suture-through-suture configuration. The repair suture **14** may be inserted through the shuttle receiving area **30**B of the shuttle suture **22**, which is generally proximate to the second end **26**. The repair suture **14** may be pulled through the shuttle suture **22** such that the distal end **16** and an adjacent section (e.g., an exposed length outside of the pledget **12**) of the repair suture **14** are on opposing sides of the shuttle suture **22**. In certain aspects, the insertion indicator **132** may be used as a guide for how far to pull the repair suture **14** through the shuttle suture **22** to maximize the efficiency of the shuttling process.

[0104] With the repair suture **14** extending through the shuttle suture **22**, the repair suture **14** may be cut (**164**) at the cut indicator **50** to remove the first needle **20** from the distal end **16**. Accordingly, the first and second needles **20**, **28** may be removed from the suture assembly **10**B for the shuttling process. For the shuttling process, the pulling force can be applied to the first end **24** of the shuttle suture **22** (**166**), which may draw the shuttle receiving area **30**B of the shuttle suture **22** and, consequently, the distal end **16** of the repair suture **14** through the replacement valve V and the tissue T. The pulling force may draw the sutures **14**, **22** into or through the pledget **12** and through the locking feature **32** (**168**).

[0105] In configurations with the spliced section **70**, the shuttle receiving area **30**B and the distal end **16** may be drawn through the spliced section **70** to pull the repair suture **14** through itself. The thickened portion of the repair suture **14** (e.g., with the stuffing **44**) proximate to the spliced section **70** may bunch, forming the locking trap to lock the repair suture **14** in the looped or stitch configuration. In configurations with the grommet **80**, the shuttle receiving area **30**B and the distal end **16** may be drawn through the grommet **80**. The inner barbs **82** may allow movement of the sutures **14**, **22** in one direction (e.g., the pulling or tensioning direction) and reduce or prevent movement in the opposing direction (e.g., a loosening direction) to lock the repair suture **14** in the stitch configuration.

[0106] The tension or pulling force may continue to be applied to the shuttle suture **22** until the shuttle receiving area **30**B and the distal end **16** are drawn through the locking feature **32**, the tissue T, and the replacement valve V (**170**) proximate to an opposing side of the pledget **12**. The shuttle suture **22** may be fully removed from the tissue T and the replacement valve V. The repair suture **14** may form the stitch or loop that extends through the tissue T and the replacement valve V three times (e.g., once from the first needle **20** and twice from the shuttling process).

[0107] The distal end **16** of the repair suture **14** may continue to be pulled to form the stitch and tension the repair suture **14** (**172**). Additional suture assemblies **10** and/or suture pairs **40** may follow this process (e.g., steps **152-172** with multiple pledgets **12** or steps **156-172** with the annular pledget **12**C) to attach the replacement valve V to the tissue T. One or more of the repair sutures **14** may be re-tensioned (**174**), which may be advantageous for initially attaching the valve V to the tissue T and then more fully forming the seal therebetween. Often, the replacement valve V may shift or move during the replacement procedure. The ability of the suture pairs **40** to be re-tensioned may allow for more accurate alignment of the valve V to be finalized once each of the suture pairs **40** has been initially tensioned. The locking features **32** may automatically lock the repair sutures **14** with each re-tensioning.

[0108] As previously noted, the pusher/cutter device may be used to apply pressure on the replacement valve V while tensioning (**172**) and re-tensioning (**174**) the repair suture **14**. This may assist with aligning the valve V and forming the fluid-tight seal. In response to the tensioning and re-tensioning, the repair sutures **14** may be cut (**176**) to be flush with the replacement valve V, removing excess length of the repair suture **14**. The locking features **32** retain the repair sutures **14** in the stitch configuration under tension. The steps **152-176** of the method **150** may be modified or conducted in different orders, with steps omitted, repeated, performed concurrently, etc. without departing from the teachings herein.

[0109] The suture assembly **10**B may provide an efficient shuttling approach with the suture-through-suture process. Additionally, the suture assembly **10**B may provide a smaller overall footprint with the shuttle suture **22** may be smaller/thinner than the shuttle suture **22** used in the suture assembly **10**A. Typically, sutures **22** that are bifurcated or have a bifurcated segment have a sufficient thickness or a sufficient number of strands/yarns for the manufacturing machine to form the two limbs. The use of the shuttle receiving area **30**B may allow for a single limb along the length of the shuttle suture **22** that is smaller/thinner than when the shuttle link **30**A is used.

[0110] With reference now to FIGS. **15** and **16**, the suture assembly **10**C is illustrated, which may function to provide the stitch configuration with a different shuttling approach compared to the suture assembly **10**A described in reference to FIGS. **1-11** and the suture assembly **10**B described in reference to FIGS. **12-14**. The suture assembly **10**C may include the pledget **12**, the repair suture **14**, the shuttle suture **22**, and the locking feature **32**. The different shuttling approach provided by the suture assembly **10**C may be based primarily on the configuration and arrangement of the shuttle suture **22**. Accordingly, the pledget **12**, the repair suture **14**, and the locking feature **32** may be substantially similar to any of the configurations described as described with reference to FIGS. **1-14**.

[0111] Similar to the prior suture assembly **10**A, **10**B configurations, in the suture assembly **10**C, the pledget **12** may support the sutures **14**, **22**. The pledget **12** is illustrated as the folded pledget **12**B but may be any configuration for supporting the sutures **14**, **22** without departing from the teachings herein. The proximal end **18** of the repair suture **14** may be coupled to the pledget **12** by the base knot **42**, and the repair suture **14** may extend through and out of the pledget **12**. The distal end **16** of the repair suture **14** may have the tapered width and may be coupled with the first needle **20** for being carried through the tissue T and the valve V. The tapered distal end **16** may assist with the shuttling process, reducing friction or resistance as the distal end **16** is drawn or pulled through the pledget **12**, the locking feature **32**, the valve V, and the tissue T.

[0112] The proximal end **18** of the repair suture **14** may have the increased thickness with the stuffing **44**, which may be additional suture material stuffed within the repair suture **14** proximate to the proximal end **18** to increase the width or thickness to assist with locking the repair suture **14**. The stuffing **44** and the thickened proximal end **18** may be configured to bunch in response to the shuttling process, forming the locking trap for retaining the repair suture **14** in the stitch configuration. As in the illustrated configuration, the locking feature **32** may be configured as the spliced section **70** with the repair suture **14** extending through the spliced section **70**.

[0113] The shuttle suture **22** may be configured to guide or shuttle the repair suture **14** through the spliced section **70** during the shuttling process, which may cause the stuffing **44** to bunch in response to tensioning or re-tensioning the repair suture **14** to lock the repair suture **14** in the stitch configuration. It is also contemplated that the grommet **80** may additionally or alternatively be used as the locking feature **32** without departing from the teachings herein. In such examples, the inner barbs **82** may allow the sutures **14**, **22** to be moved through the locking feature **32** in the pulling/tensioning direction and reduce or prevent movement of the sutures **14**, **22** in the opposing direction to lock the tensioned repair suture **14** in the stitch configuration.

[0114] In the suture assembly **10**C illustrated in FIGS. **15** and **16**, the shuttle suture **22** may be folded to form the shuttling feature **30**, which may be configured as the folded shuttle loop **30**C for use in the shuttling process. In certain aspects, the shuttle suture **22** may be folded at a general midpoint, mid-region, or mid-section **200** between the first and second ends **24**, **26** to form two limbs. The folded shuttle suture **22** may extend through the pledget **12**.

[0115] Accordingly, in this configuration, the first and second ends **24**, **26** of the shuttle suture **22** may be disposed adjacent to one another, and the ends **24**, **26** may be spaced from the mid-section **200** by the pledget **12**. In this regard, both limbs of the folded shuttle suture **22** may extend through the pledget **12**, with a first limb extending between the first end **24** and the mid-section **200** and the second limb extending between the second end **26** and the mid-section **200**. A section of the first limb between the first end **24** and the mid-section **200** may extend through the pledget **12** adjacent to a section of the second limb between the second end **26** and the mid-section **200**. Generally, both limbs may engage the pledget **12** and/or the locking feature **32** in a same or similar location for shuttling or guiding the repair suture **14** and for removing the shuttle suture **22** from the pledget **12** after the stitch is formed.

[0116] To utilize the suture assembly **10**C to couple the implant or valve V with the tissue T and form the fluid-tight seal, the sutures **14**, **22** may be coupled with the needles **20**, **28** for carrying the sutures **14**, **22** through the tissue T and the valve V. In certain aspects, the first and second end **24**, **26** of the shuttle suture **22** may be disposed adjacent to the distal end **16** of the repair suture **14**. Each of the ends **18**, **24**, **26** may be coupled with the first needle **20** to allow the ends **18**, **24**, **26** to be carried through the tissue T and the valve V concurrently. The ends **18**, **24**, **26** may be separately coupled to the needle **20** or may combined or spliced together to reduce the number of tails coupled to the needle **20**.

[0117] The mid-section **200** of the shuttle suture **22** may be carried through the tissue T and the valve V separately from the ends **24**, **26**, generally using the second needle **28**. The folded shuttle loop **30**C may be coupled with the second needle **28** via a connecting suture **202**. The connecting suture **202** may form a loop through the folded shuttle loop **30**C and have both ends or tails coupled to the second needle **28**, consequently coupling the folded shuttle loop **30**C with the second needle **28**. The connecting suture **202** may be constructed similarly to the repair suture **14** and/or the shuttle suture **22**. The second needle **28** may carry the connecting suture **202** and the folded shuttle loop **30**C through the tissue T and the valve V. The needles **20**, **28** may be utilized to arrange the pledget **12** adjacent to the inner surface of the tissue T with the ends **18**, **24**, **26** and the mid-section **200** on an opposing side of the tissue T and the valve V.

[0118] With the ends **18**, **24**, **26** and the mid-section **200** carried through the tissue T and the valve V, the needles **20**, **28** may be removed for the shuttling process. The folded shuttle loop **30**C may be formed by the two limbs between the pledget **12** and the mid-section **200** (e.g., the middle section of the shuttle suture **22** forms the folded shuttle loop **30**C). This may provide a larger insertion location for the repair suture **14** for the shuttling process. The distal end **16** of the repair suture **14** may be inserted through the folded shuttle loop **30**C and folded, such as at the fold indicator **56**. This engagement may allow the shuttle suture **22** to carry or shuttle the repair suture **14** to form the stitch.

[0119] Tension may be applied to both ends **24**, **26** of the shuttle suture **22**. In response, the folded shuttle loop **30**C can be drawn through the pledget **12** and the locking feature **32**. As the folded shuttle loop **30**C is pulled by the tension or pulling force, the folded shuttle loop **30**C may also pull the distal end **16** of the repair suture **14** through the pledget **12** and the locking feature **32**. The locking feature **32** may be configured to lock the repair suture **14** in the looped or stitch configuration, such as by bunching the stuffing **44** when the locking feature **32** is the spliced section **70** or with the inner barbs **82** when the locking feature **32** is the grommet **80**. In the stitch configuration, the full extension of the repair suture **14** may be: from the base knot **42**, through the pledget **12**, through the tissue T, through the valve V, across the valve V, through the valve V, through the tissue T, through the locking feature **32**, through the tissue T, and through the valve V. Accordingly, the repair suture **14** may attach and form a seal between the valve V to the tissue T.

[0120] The folded suture **22** may result in two limbs of the shuttle suture **22** being carried through the tissue T and valve V concurrently. To utilize the folded shuttle suture **22**, the shuttle suture **22** may be a smaller suture compared to at least the shuttle suture **22** with the bifurcated shuttle link **30**A in the suture assembly **10**A (see FIGS. **3-10**). It is also contemplated that the folded shuttle suture **22** in this configuration of the suture assembly **10**C may be a smaller suture **22** than the shuttle suture **22** with the shuttle receiving area **30**B in the suture assembly **10**B (see FIGS. **12** and **13**). The smaller shuttle suture **22** in the suture assembly **10**C may allow the shuttle suture **22** to be folded while maintaining a low profile or smaller overall suture assembly **10**C. In certain aspects, the shuttle suture **22** may be a monofilament suture. Additionally or alternatively, the shuttle suture **22** may be a 4-0 or 5-0 suture, such as, for example, a 4-0 or 5-0 Prolene® suture. The smaller shuttle suture **22** may allow the suture **22** to be folded to form the folded shuttle loop **30**C and to reduce the overall footprint of the suture assembly **10**C.

[0121] The configuration and size of the shuttle suture **22** may assist with providing a large enough shuttle suture **22** to receive and guide the repair suture **14** while maintaining a low profile overall assembly **10**. The folded shuttle suture **22** may also increase manufacturing efficiency. For example, forming bifurcated sutures **22** can increase complexity in the manufacturing process, as well as increase the size of the shuttle suture **22** being used. A certain number of yarns or strands of material may typically be utilized to form bifurcated suture segments, which can increase the size of the suture **22**.
[0122] Referring to FIG. **17**, as well as FIGS. **14** and **15**, the suture assembly **10**C may form the stitch and can be used in a valve replacement procedure or other implant attachment procedure in a substantially similar manner as the suture assemblies **10**A, **10**B, with differences primarily in the shuttling approach and how the shuttle suture **22** is inserted through the tissue T and the valve V with the needles **20**, **28**. A method **220** of using the suture assembly **10**C for a valve replacement may include providing the suture assembly **10**C (**222**) that includes the pledget **12**, at least one repair suture **14** and associated shuttle suture **22**, and the two needles **20**, **28**. As previously noted, the leaflets may be removed from the patient, and the replacement valve V may be provided. The pledget **12** can be arranged on the underside of the annulus A (**224**).
[0123] The first needle **20** can be inserted through the tissue T and the replacement valve V (**226**). The first needle **20** may carry the distal end **16** of the repair suture **14** and both the first end **24** and the second end **26** of the shuttle suture **22** through the tissue T and the replacement valve V. Similarly, the second needle **28** may be inserted through the tissue T and the replacement valve V (**228**). The second needle **28** may carry the connecting suture **202** and the folded shuttle loop **30**C of the shuttle suture **22** through the tissue T and the replacement valve V. At this stage, the ends **18**, **24**, **26** of the sutures **14**, **22**, the folded shuttle loop **30**C (e.g., the mid-section **200** of the shuttle suture **22**), and the needles **20**, **28** may be on the opposing side of the tissue T and the replacement valve V compared to the pledget **12**. Further, the ends **18**, **24**, **26** of the sutures **14**, **22** may generally be spaced from the folded shuttle loop **30**C, which can allow the repair suture **14** to extend over the valve V to form the stitch as a result of the shuttling process. Tension may be applied by pulling both needles **20**, **28** to move the pledget **12** to engage the tissue T and reduce excess suture **14**, **22** length under the tissue T and/or valve V.
[0124] With the ends **18**, **24**, **26** and the folded shuttle loop **30**C carried through the tissue T and the valve V, the sutures **14**, **22** may be cut (**230**) at the cut indicators **50-54** to remove the first needle **20** from the ends **18**, **24**, **26** of the sutures **14**, **22** for the shuttling process. With the removal of the first needle **20**, the distal end **16** of the repair suture **14** and the ends **24**, **26** of the shuttle suture **22** may be separated from one another. The connecting suture **202** may be cut (**232**) to remove the second needle **28** and the connecting suture **202** from the folded shuttle loop **30**C of the shuttle suture **22**.
[0125] The distal end **16** of the repair suture **14** may be inserted or threaded through the folded shuttle loop **30**C (**234**) for the shuttling process. The distal end **16** may be folded at the fold indicator(s) **56** to form the folded distal end **16** around the folded shuttle loop **30**C. The pulling force is applied to the ends **24**, **26** of the shuttle suture **22** (**236**), which may draw the folded shuttle loop **30**C with the folded distal end **16** of the repair suture **14** through the replacement valve V and the tissue T. The pulling force may draw the sutures **14**, **22** into or through the pledget **12** and through the locking feature **32** (**238**). In certain aspects, the folded shuttle loop **30**C and the folded distal end **16** may be drawn through the spliced section **70** to pull the repair suture **14** through itself, bunching the stuffing **44** and forming the locking trap to lock the repair suture **14** in the stitch configuration. Additionally or alternatively, the folded shuttle loop **30**C and the folded distal end **16** may be drawn through the grommet **80**, with the inner barbs **82** locking the repair suture **14**.
[0126] The pulling force may continue to be applied to the shuttle suture **22** until the folded shuttle loop **30**C and the folded distal end **16** are drawn through the tissue T and the replacement valve V (**240**) proximate to an opposing side of the pledget **12** (e.g., after being pulled through the locking

feature **32**). With the shuttling process, the shuttle suture **22** may be fully removed from the tissue T and the replacement valve V. Further, with the shuttle suture **22** removed, the repair suture **14** generally remains in the stitch or loop configuration, connecting the valve V to the tissue T.

[0127] To form the seal between the tissue T and the valve V, the repair suture **14** may continue to be pulled to form the stitch and tension the repair suture **14** (**242**). Additional suture assemblies **10** and/or suture pairs **40** may follow this process (e.g., steps **222-242** with multiple pledgets **12** or steps **226-242** with the annular pledget **12**C) to attach the replacement valve V to the tissue T. As the suture pairs **40** are tensioned, the valve V may move in response to the tension forces being applied. The valve V may be moved after an initial tensioning process, and one or more of the repair sutures **14** may be re-tensioned (**244**) to more fully form the seal between the tissue T and the valve V. Additional pulling force may be applied to the distal ends **18** of the repair sutures **14** to re-tension the repair sutures **14**, and the locking features **32** may automatically lock the repair sutures **14** with each re-tensioning.

[0128] As previously noted, the pusher/cutter device may be used to apply pressure on the replacement valve V while tensioning (**242**) and re-tensioning (**244**) the repair suture **14**. This may be advantageous for reducing displacement of the replacement valve V and reducing slack from the suture assemblies **10**C. The repair suture **14** may be cut (**246**) to be flush with the replacement valve V, removing excess length from the repair suture **14** from the surgical site. The locking features **32** retain the repair sutures **14** in the stitch configuration under tension. The steps **222-246** of the method **220** may be modified or conducted in different orders, with steps omitted, repeated, performed concurrently, etc. without departing from the teachings herein.

[0129] Referring again to FIGS. **3-17**, the suture assemblies **10**A, **10**B, **10**C may include the repair suture **14** fixed to the pledget **12** and the shuttle suture **22**. Each of the shuttle suture **22** and the repair suture **14** may be coupled to one or both of the needles **20**, **28** for carrying the suture assembly **10** through the tissue T and the replacement valve V or other implants. Accordingly, with a single pass-through of each needle **20**, **28**, one or both of the repair suture **14** and the shuttle suture **22** may be carried through the tissue T and the valve V to position the pledget **12** against the tissue T and make the ends **16**, **18**, **24**, **26** available on an outer side of the valve V for the shuttling process.

[0130] With reference to FIGS. **18-22**, additional exemplary configurations of the knotless suture assembly **10** are illustrated, including suture assemblies **10**D, **10**E. The suture assemblies **10**D, **10**E may utilize the repair suture **14** and the shuttle suture **22**. The repair suture **14** may be utilized to attach implants, such as the replacement valve V, to the tissue T, as described with respect to FIGS. **1-17**, or to connect multiple tissue bodies T**1**, T**2** together, as will be described further herein with respect to FIGS. **25-38**. The shuttle suture **22** may be utilized to shuttle or guide the repair suture **14** to form the stitch configuration in response to the pulling force or tension applied to the shuttle suture **22**, similar to the process described above with respect to FIGS. **4-6**. Sections of the repair suture **14** and the shuttle suture **22** may be spliced or otherwise combined together during a manufacturing process for efficient insertion during a surgical procedure and may be configured to be at least partially separated during the procedure to proceed with the shuttling process and subsequent removal of the shuttle suture **22**.

[0131] Referring to FIGS. **18** and **19**, in this configuration of the suture assembly **10**D, the suture assembly **10**D may include the repair suture **14** having a substantially similar length as the shuttle suture **22**. The first end **16** of the repair suture **14** may be arranged adjacent to the first end **24** of the shuttle suture **22**, and the second end **18** of the repair suture **14** may be disposed adjacent to the second end **26** of the shuttle suture **22**. In other words, the second end **18** of the repair suture **14** may not be fixed.

[0132] The suture assembly **10**D may include one or more interlacing or spliced segments **252**, **254**, **256** that couple the repair suture **14** and the shuttle suture **22**. At these segments **252**, **254**, **256** the shuttle suture **22** may be spliced into, combined with, embedded in, encapsulated in, interlaced with, or otherwise joined with the repair suture **14**. Typically, the shuttle suture **22** may be spliced into the repair suture **14**; however, the repair suture **14** may be spliced into the shuttle suture **22** without departing from the teachings herein. The spliced segments **252**, **254**, **256** may allow the sutures **14**, **22** to be interlaced together forming an integrated suture assembly **10**D that can maximize the efficiency of the procedure. For example, the integrated suture assembly **10**D may be inserted into/through the tissue T as a single body, and the sutures **14**, **22** may then be at least partially separated to move independently of one another for the shuttling process.

[0133] The repair suture **14** may include the first end **16**, which may be referred to herein as a lead or movable end **16**, and the second opposing end **18**, which may be referred to herein as a static end **18**. The repair suture **14** may have a predefined length between the first and second ends **16**, **18**. The predefined length may depend on the patient anatomy or procedure for which the suture assembly **10** is being used. One or both of the ends **16**, **18** may be tapered to assist with inserting the suture assembly **10** through the tissue T. The shuttle suture **22** may include the first end **24**, which may be referred to as the non-looped end **24**, and the second opposing end **26**, which may be referred to as a looped end **26**. The looped end **26** may be arranged proximate to the static end **18** of the repair suture **14**, and the non-looped end **24** may be arranged proximate to the lead end **16** of the repair suture **14**.

[0134] The second end **26** of the shuttle suture **22** may define or include the end shuttle loop **30**D, which may be an opening formed from folding the suture **22**. The end **26** of the shuttle suture **22** may be folded to engage itself to form the end loop **30**D. In this configuration, the width of the shuttle suture **22** at the end loop **30**D may be greater than the width of the remainder of the shuttle suture **22**. Alternatively, the end **26** may be tapered to reduce the thickness of the loop **30**D. This configuration may be advantageous for maximizing the efficiency of the manufacturing process. The shuttle loop **30**D may also be formed through a bifurcation process. In bifurcated examples, during the manufacturing process, the shuttle suture **22** may be formed of interlacing strands as a single limb, then formed into two limbs to form the bifurcated segment, and then again formed as a single limb to form the shutting feature **30**, similar to the shuttle link **30**A described with respect to FIGS. **3-8**. This process may be advantageous for retaining a smaller width of the shuttle suture **22** at the shuttle end loop **30**D.

[0135] The repair suture **14** and the shuttle suture **22** may be spliced or otherwise combined together in segments **252**, **254**, **256** at multiple locations along the length of the repair suture **14** to form the combined or integrated suture assembly **10**D. The suture assembly **10**D may be manufactured to have the preformed spliced segments **252**, **254**, **256** and allow the integrated suture assembly **10** to be readily unpackaged and used in a sterile setting. The preformed engagement or spliced segments **252**, **254**, **256** may be advantageous for maximizing the efficiency of the procedure and reducing separate lengths of the sutures **14**, **22** that may become tangled during the procedure.

[0136] At the first spliced segment **252**, the first end **24** of the shuttle suture **22** may be spliced into the repair suture **14** at or proximate to the first end **16** of the repair suture **14** (e.g., at a first location). In certain aspects, the repair suture **14** may extend a predefined length beyond the first end **24** of the shuttle suture **22**. The predefined length may be minimal such that the ends **16**, **24** may be adjacent to one another or may be greater such that a portion of the repair suture **14** extends beyond the end **24** of the shuttle suture **22** that is combined, joined, embedded, or otherwise spliced into the repair suture **14**. By splicing the end **24** of the shuttle suture **22** into the repair suture **14**, the end of the suture assembly **10** may have a single tail.

[0137] The shuttle suture **22** may be spliced into the repair suture **14** at a second location forming the second spliced segment **254**, t. For example, the second end **26** of the shuttle suture **22** may be spliced into the repair suture **14** at or proximate to the second end **18**. Accordingly, the shuttle loop **30**D may be spliced into the repair suture **14**. Similar to the first ends **16**, **24**, the repair suture **14** may extend a predefined length beyond the second end **26** of the shuttle suture **22**. This predefined length may be minimal such that the ends **18**, **26** may be adjacent to one another or may be greater such that a portion of the repair suture **14** extends beyond the end **26** of the shuttle suture **22** that is combined, joined, embedded, or otherwise spliced into the repair suture **14**. By splicing the end **26** of the shuttle suture **22** into the repair suture **14**, both ends of the suture assembly **10** may have a single tail, which can subsequently be separated after insertion through the tissue T for the shuttling process.

[0138] The shuttle suture **22** may be spliced into the repair suture **14** at a third location. The third spliced segment **256** may be between the first and second spliced segments **252**, **254** and may be generally aligned with the mid-section **200** and a mid-region **258** of one or both of the repair suture **14** and the shuttle suture **22**. The third spliced segment **256** may form the locking feature **32** of the suture assembly **10** (similar to the spliced section **70** described above). The third spliced segment **256** may allow the shuttle suture **22** to move relative to the repair suture **14** to, ultimately, move through and out of the repair suture **14** during the shuttling process.

[0139] In various aspects, additional stuffing **44** may be spliced or otherwise integrated into the repair suture **14**, which may be at a spliced region **262**. In the illustrated example, the stuffing **44**, which may also be referred to as a stuffing suture **44**, may be added at the spliced region **262** between the first spliced segment **252** and the third spliced segment **256**. Typically, the stuffing **44** may be adjacent to the third spliced segment **256** (e.g., the locking feature **32**) on an opposite side of the shuttle loop **30**D. During the shuttling process, the repair suture **14** may be drawn through the third spliced segment **256** and toward the stuffing **44**, which may cause the stuffing **44** to bunch and lock the repair suture **14** in the looped, stitch configuration.

[0140] The ends of the suture assembly **10**D may be coupled with the needles **20**, **28**, respectively. As the shuttle suture **22** may be spliced into the repair suture **14**, the first and second ends **16**, **18** of the repair suture **14** may be coupled with the needles **20**, **28**, respectively, which can carry the ends **24**, **26** of the shuttle suture **22** through the tissue T. Accordingly, the suture assembly **10**D may include the two needles **20**, **28** for carrying the suture assembly **10**D through the tissue T and/or the implant/valve V in a manner substantially similar as described with respect to FIGS. **3-8**. The ends **24**, **26** of the shuttle suture **22** being spliced into the repair suture **14** may reduce resistance or friction as the suture assembly **10**D is carried through the tissue T and/or valve V. Additionally, a single suture body (e.g., the repair suture **14** or the combined repair suture **14** and shuttle suture **22**) may initially be inserted through the tissue T with the needles **20**, **28**, which may assist with a smoother insertion process.

[0141] After the needles **20**, **28** have carried the suture assembly **10** through the tissue T and/or the valve V, the ends of the suture assembly **10** may be cut. The suture assembly **10**D may have cut indicators **52**, **54** to identify an optimal location for cutting the suture assembly **10**D. Cutting the suture assembly **10**D may separate the ends **16**, **18** of the repair suture **14** from the ends **24**, **26** of the shuttle suture **22**. Accordingly, the suture assembly **10**D may be cut between the middle spliced segment **256** and each of the end spliced segments **252**, **254**. Alternatively, cutting at the spliced segments **252**, **254** may allow the sutures **14**, **22** to be separated.

[0142] Accordingly, the sutures **14**, **22** may be joined or spliced together for insertion through the tissue T and then separated with the removal of the needles **20**, **28** for the shuttling process. In this way, cutting the needles **20**, **28** from the suture assembly **10** may expose the shuttle loop **30**D and separate the movable end **16** of the repair suture **14** from the non-looped end **24** of the shuttle suture **22**. The movable end **16** may be inserted through the shuttle loop **30**D, folded, and carried through the middle spliced segment **256**, similar to the process described with respect to FIG. **4-6**. The locking feature **32**, which may include the third spliced segment **256** and the stuffing **44**, may lock the repair suture **14** in the stitch configuration.

[0143] As illustrated in FIGS. **18** and **19**, the suture assembly **10**D may not include the pledget **12**. Such configurations may be advantageous for more orthopedic approaches and procedures, such as connecting tissue bodies T**1**, T**2** together. In additional or alternative exampled, such as illustrated in FIG. **20**, the suture assembly **10**D may include the pledget **12**. The suture assembly **10**D may exit from different points on the pledget **12** to create a buttress for anchoring in the tissue T. In certain aspects, the repair suture **14** and the shuttle suture **22** may each exit the pledget **12** at the same two opposing points or at two different opposing points. The suture assembly **10**D may extend from opposing sides of the pledget **12**, which may have any configuration described herein. The pledget **12** may assist in distributing the force applied by the suture assembly **10**D.

[0144] Referring now to FIGS. **21-23**, the suture assembly **10**E is illustrated, which may be similar to the suture assembly **10**D having multiple spliced segments **252**, **254** between the repair suture **14** and the shuttle suture **22**. The suture assembly **10**E may include two spliced segments **252**, **254** between the repair suture **14** and the shuttle suture **22** and may omit the third spliced segment **256**. Additionally, at least one end **26** of the shuttle suture **22** may not be included in the spliced segments **252**, **254**. In this illustrated example, the non-looped end **24** may be included in the spliced segment **252** while the shuttle loop **30**D may not be included in a spliced segment **254** (e.g., the shuttle loop **30**D may be exposed and spaced from the spliced segment **254**). The repair suture **14** may be substantially similar as in the suture assembly **10**D or may have different aspects to form and locate the spliced segments **252**, **254**.

[0145] Values or measurements are included herein and are understood to be merely exemplary to illustrate the relationship between various sections of the sutures **14**, **22** and are, therefore, not limiting. In various non-limiting aspects, the repair suture **14** may be provided and may be between about 30 cm and about 40 cm, such as, for example, about 36 cm. On a first side of the mid-region **258** of the repair suture **14** that includes the first end **16**, the repair suture **14** may include three limbs **264**, **266**, **268**. At least one of the limbs **264**, **266** may be about one-half the overall length of the repair suture **14**, such as about 18 cm. The second limb **266** may be the same length as the first limb **264** or may be shorter.

[0146] The repair suture **14** may include a transition section **270** where the two limbs **264**, **266** merge into a single limb, and the single limb section may be referred to as the mid-region **258** of the repair suture **14**. The mid-region **258** may be at a centerline of the repair suture **14** or may be closer to one of the ends **16**, **18** without departing from the teachings herein. The transition section **270** may be about 1 cm in length, which may be included in the length of the one or both limbs **264**, **266**.

[0147] The first limb **264** may further form or extend into an additional third limb **268** at an additional transition section **272**. Accordingly, on the first side of the mid-region **258** forming the first end **16**, the repair suture **14** may include three limbs **264**, **266**, **268** with the third limb **268** formed from or coupled with one of the other limbs **264** and having a shorter length. The transition section **272** may be between about 0.2 cm to about 0.5 cm and may extend along the length of the repair suture **14**, which may form a portion of the length of the first and third limbs **264**, **268**. The third limb **268** may be about 10 cm in length.

[0148] Referring still to FIGS. **21-24**, on an opposing side of the mid-region **258** compared to the first end **16** (e.g., the side including the second end **18**), the repair suture **14** may form or include two limbs **276**, **278**. The limbs **276**, **278** may be between about one-half and one-quarter of the length of the repair suture **14**. For example, the limbs **276**, **278** may be about one-third of the overall length of the repair suture **14**, such as about 12 cm. The two limbs **276**, **278** on the second side of the mid-region **258** may be shorter than the two longer limbs **264**, **266** on the first side of the mid-region **258**. On the second side of the mid-region **258** forming the second end **18** of the repair suture **14**, the repair suture **14** may have a third transition section **282** where the repair suture **14** can merge from the two limbs **276**, **278** and into the single limb mid-region **258**. The transition section **282** may be between about 2 mm and about 6 mm along the length of the repair suture **14**, which may be included in the length of the two limbs **276**, **278**.

[0149] The transition sections **270**, **272**, **282** may be spliced sections of the repair suture **14** where two limbs **264**, **266**, **268**, **276**, **278** merge or combine into a single limb, respectively. The multilimb **264**, **266**, **268**, **276**, **278** sections of the repair suture **14** may be formed as bifurcated segments. In such examples, during the manufacturing process of the repair suture **14**, the strands may be carried in pattern(s) to: form the two limbs **276**, **278**, be spliced together at the transition section **282** to form the single limb mid-region **258**, again be bifurcated to form two limbs **276**, **278** on the opposing side of the mid-region **258** and at the transition section **270**, and the first limb **264** may be further bifurcated to form the third limb **268** at the transition section **272**. The bifurcated limbs **264**, **266**, **268**, **276**, **278** of the repair suture **14** may have a lesser width than the single limb mid-region **258**, which may reduce the overall size of the suture assembly **10** when the repair suture **14** is combined with the shuttle suture **22**.

[0150] Alternatively, the different limbs **264**, **266**, **268**, **276**, **278** may be manufactured separately. The limbs **264**, **266**, **268**, **276**, **278** may then be spliced together at the transition sections **270**, **272**, **282**, respectively, to combine into single limb segments. The repair suture **14** may also be manufactured through a combination of bifurcated segments and limb segments that are spliced together.

[0151] Referring still to FIGS. **21-23**, the transition sections **270**, **272**, **282** may provide locations for forming the spliced segments **252**, **254** between the repair suture **14** and the shuttle suture **22**. The shuttle suture **22** may be arranged proximate to the repair suture **14** to be combined or spliced into the repair suture **14** at one or more locations. Generally, the shuttle end loop **30**D may be disposed proximate to the static end **18**, and the non-looped end **24** may be disposed proximate to the lead end **16** with the three limbs **264**, **266**, **268**.

[0152] The non-looped end **24** of the shuttle suture **22** may be spliced with the lead end **16** of the repair suture **14**. The first and second transition sections **270**, **272** may form the boundaries for the first spliced segment **252** between the repair suture **14** and the shuttle suture **22**. In various aspects, the first end **24** of the shuttle suture **22** may be spliced into the repair suture **14** between the transition sections **270**, **272**. The shuttle suture **22** may be arranged at or adjacent to the proximal transition section **270** and/or at or adjacent to the distal transition section **272**. Additionally or alternatively, the shuttle suture **22** may be spaced from one or both of the transition sections **270**, **272**.

[0153] The spliced segment **252** may be between about 4 cm and about 5 cm in length. The spliced segment **252** may be spaced between about 1 cm and about 2 cm from the single limb mid-region **258**. This spacing may align with an end of the first transition section **270**. The shuttle suture **22** may be spliced into the first limb **264** at the end of the first transition section **270** and extend toward the second transition section **272**. The end **24** of the shuttle suture **22** may be substantially or fully encapsulated in the repair suture **14**, such that the end **24** of the shuttle suture **22** may be part of the spliced segment **252**. In this way, the repair suture **14** may extend beyond the first end **24** of the shuttle suture **22**.

[0154] Additionally, the shuttle suture **22** may be spliced into repair suture **14** proximate to the third transition section **282** to form the second spliced segment **254**. In the illustrated example, the shuttle suture **22** may be spliced into the first limb **276** in a manner that minimizes the distance between the spliced segment **254** and the end of the spliced limb **276** (e.g., close to where the limbs **276**, **278** are spliced or combined at the transition section **282**). In non-limiting aspects, the maximum distance between the spliced segment **254** and the merge location of the two limbs **276**, **278** may be about 1 mm. The spliced segment **254** may be between about 5 mm and about 10 mm along the length of the sutures **14**, **22**. The shuttle suture **22** may be interlaced with the repair suture **14** and/or strands thereof to form the combined spliced segments **252**, **254** between the sutures **14**, **22**.

[0155] Referring still to FIGS. **20-22**, the mid-region **258** of the repair suture **14** may be disposed between the first and third spliced segments **252**, **256** formed between the repair suture **14** and the shuttle suture **22**. The mid-region **258** may have an increased or greater thickness compared to the ends **16**, **18** of the repair suture **14** from being formed as a single limb and may also have a further increased thickness to form the locking feature **32** for the knotless stitch configuration. The repair suture **14** may include the stuffing **44**, which may also be referred to as the stuffing suture **44**. The stuffing **44** may be additional suture material stuffed within or interlaced with the repair suture **14** to increase the width or thickness and assist in locking the repair suture **14** in the stitch configuration. The stuffing suture **44** may be spliced into the repair suture **14** at the mid-region **258**, such as by being inserted into, woven, braided, or otherwise interlaced with the repair suture **14** during the manufacturing process.

[0156] The stuffing suture **44** may be spliced into the repair suture **14** at the additional spliced region **262** between the spliced segments **252**, **256** with the shuttle suture **22**. The thickened portion formed by the stuffing suture **44** may abut or be spaced from one or both of the spliced segments **252**, **256**. In the illustrated configuration, the stuffing suture **44** may be spaced from both spliced segments **252**, **256** and may be closer to the third spliced segment **256** than the first segment **252**. The stuffing suture **44** may form the spliced region **262** at the mid-region **258** that extends between about 3 cm and about 5 cm along the length of the repair suture **14**.

[0157] The stuffing suture **44** may be substantially or fully encapsulated in the repair suture **14** to increase the thickness of the repair suture **14** and reduce any excess material extending from the repair suture **14**. The stuffing suture **44** may be arranged off-center in the single limb mid-region **258**. As illustrated, the stuffing suture **44** may be closer to a second end of the mid-region **258** (proximate to the third spliced segment **256** on the side with the loop **30**D) than a first end (on a side with the non-looped end **24**). This may be advantageous for bunching the stuffing **44** to lock the repair suture **14** during the shuttling or tensioning processes while allowing some movement of the stuffing suture **44** within the mid-region **258** and at least substantially containing the stuffing **44** within the mid-region **258**. In this configuration, with the third spliced segment **256** (see FIGS. **18-20**) being omitted, the second spliced segment **254** may operate as the locking feature **32**.

[0158] Referring again to FIGS. **21-23**, the three limbs **264**, **266**, **268** at the first end may be combined or joined to form the first end **16** of the repair suture **14**, and the two limbs **276**, **278** may be combined or joined to form the second end **18** of the repair suture **14**. Combining the limbs **264**, **266**, **268**, **276**, **278** may result in the repair suture **14** being a single limb along the length thereof. The limbs **264**, **266**, **268**, **276**, **278** may be combined via a variety of processes, such as interlacing, weaving, braiding, adhesives, etc. One or both of the ends **16**, **18** of the repair suture **14** may be tapered, having a reduced or lesser width/thickness. The tapering may occur during the manufacturing process.

[0159] For example, in aspects where the repair suture **14** is formed of interlacing strands, fewer strands of material may be used to form the tapered ends **16**, **18**. The tapering may also be accomplished by tapering the ends of each limb **264**, **266**, **268**, **276**, **278** to collectively form tapered ends **16**, **18** of the repair suture **14**. The tapered ends **16**, **18** may be about 1 cm in length and may assist with reducing friction/resistance during insertion of the suture assembly **10** into/through the tissue T and during the shuttling process. It is contemplated that the lead end **16** of the repair suture **14** may be tapered while the static end **18** may not be as the lead end **16** is inserted through the tissue T. In examples where both the first and second ends **16**, **18** are movable or lead ends, both ends **16**, **18** may be tapered.

[0160] As illustrated in FIGS. **22** and **23**, the second spliced segment **254** may be a discrete segment spaced from the ends **18**, **26** of the sutures **14**, **22**. Accordingly, one end of the suture assembly **10**E may be a combined end, which may be used for carrying the suture assembly **10**E through the tissue T and/or the implant. The non-combined end including the shuttle loop **30**D may not be actively inserted through the tissue T prior to the shuttling process. This configuration may leave the shuttle loop **30**D as a loose or tail end separated from the repair suture **14**.

[0161] In certain aspects, the shuttle end loop **30**D in the suture assembly **10**E may be spliced into and at least partially encapsulated in the repair suture **14**, such as in the configuration illustrated in FIGS. **18-20**. In such aspects, both ends **24**, **26** of the shuttle suture **22** may be spliced into the repair suture **14**. This may provide a suture assembly **10** with two combined ends that may be passed through tissue T and/or a valve V in the combined state and then separated for the shuttling process. This may be advantageous for maximizing the efficiency of the insertion process and minimizing the number of separate sutures **14**, **22** passing through the tissue T. Moreover, the combined or spliced state of the suture assembly **10** may result in less friction or resistance in passing the suture assembly **10** through the tissue T.

[0162] The suture assembly **10**E may be coupled with one or more needles **20**, **28** for passing the suture assembly **10** through the tissue T and/or the valve V. In examples where the non-looped end **24** is spliced into the repair suture **14** while the looped end **26** is not, a single needle **20** may be coupled to the one or both of the second or lead ends **16**, **24** (e.g., a combined lead end of the suture assembly **10**E) to pass the suture assembly **10** through the tissue T. In this way, the lead end of the suture assembly **10**E coupled with the needle **20** may be passed through the tissue T at multiple locations to position the opposing ends **18**, **26** on a same side of the tissue T without passing the ends **18**, **26** through the tissue T. Where the end **24** of the shuttle suture **22** is encapsulated in the repair suture **14** such that repair suture **14** extends beyond the end **24** of the shuttle suture **22**, at least the lead end **16** of the repair suture **14** may be coupled to the needle **20** and used to insert or carry the suture assembly **10** through the tissue T.

[0163] In examples where both ends **24**, **26** of the shuttle suture **22** are spliced into the repair suture **14**, at least one of the first ends **16**, **24** and at least one of the second ends **18**, **26** of the suture assembly **10** may be coupled with a respective needle **20**, **28**, similar to the configuration illustrated in FIGS. **18-20**. In such examples, one or both ends of the suture assembly **10** may be passed through the tissue T using the respective needle **20**, **28**. In additional non-limiting examples, the suture assembly **10** may not include or be coupled with the needle(s) **20**, **28**. In such examples, the suture assembly **10** may be carried or inserted through tissue T using a device, such as a suture passer or implant device as described further herein.

[0164] Referring still to FIGS. **21-23**, the suture assembly **10**E may not include the pledget **12**. This may be advantageous for orthopedic procedures, such as attaching two tissue bodies T**1**, T**2** as described herein with respect to FIGS. **34-37**. However, the suture assembly **10**E may include the pledget **12** without departing from the teachings herein, such as when the suture assembly **10**E is used for valve replacement and other implant attachment procedures. In such configurations, the spliced segment **254** between the repair suture **14** and the shuttle suture **22** forming the locking feature **32** may be arranged in or adjacent to the pledget **12** to couple to the suture assembly **10**E with the pledget **12**, similar to the configurations illustrated in FIGS. **4-8**, **10-12**, **15**, **16**, and **20**. The pledget **12** may assist with distributing pressure applied by the repair suture **14**. The suture assembly **10** may exit the pledget **12** at different locations to create a buttress point for anchoring in tissue T.

[0165] Referring to FIG. **24**, as well as FIGS. **18-23**, the suture assembly **10** may be manufactured to combine the repair suture **14** with the shuttle suture **22** at one or more of the spliced segments **252**, **254**, **256** and the stuffing suture **44** at the spliced region **262**. A method (**310**) of manufacturing the suture assembly **10** may include trimming or cutting strands of suture material to the predefined length(s) (**312**). The repair suture **14** may be formed (**314**). The repair suture **14** may be formed as a single limb or with multiple limbs **264**, **266**, **268**, **276**, **278** (**314**). The repair suture **14** may be formed during a single process where the strands can be arranged in bifurcated segments and spliced together to form the repair suture **14** with the multiple limbs **264**, **266**, **268**, **276**, **278** on each end **16**, **18**, respectively.

[0166] Alternatively, different limbs or segments of suture material may be formed initially, and the segments may be spliced together to form the transition sections **270**, **272**, **282** and, ultimately, form the repair suture **14** with the mid-region **258** and multiple limbs **264**, **266**, **268**, **276**, **278**. The suture material may be spliced together to form alternative limbs **264**, **266**, **268**, **276**, **278** at specified lengths. The limbs **264**, **266**, **268**, **276**, **278** may be trimmed to reduce or prevent fraying and the ends thereof. Trimming the limbs **264**, **266**, **268**, **276**, **278** may also ensure that the shuttle suture **22** is substantially or fully encapsulated.

[0167] The shuttle suture **22** may be provided (**316**). The shuttle suture **22** may be spliced into the repair suture **14** (**318**) and form the first spliced segment **252** (**320**). The first spliced segment **252** may include the first end **24** of the shuttle suture **22** and may include or be adjacent to the first end **16** of the repair suture **14**. The first spliced segment **252** may be formed to be adjacent to the end of the spliced limb **264** proximate to the transition section **270**. In certain aspects, the shuttle suture **22** may be spliced into the first limb **264** on the first side of the mid-region **258**, which may be the limb **264** that includes the second transition section **272** to form the third limb **268**. This spliced segment **254** may be between the two transition sections **272**, **282** on the first side of the mid-region **258**.

[0168] The stuffing **44** or stuffing suture **44** may be spliced into the repair suture **14** at the mid-region **258** (**322**). The stuffing suture **44** may be centered or off-center within the mid-region **258**. The spliced region **262** formed between the repair suture **14** and the stuffing **44** and the stuffing **44** may extend along the length of the mid-region **258**, such as between the first transition section **270** and the third transition section **282**. The spliced region **262** may be formed between a centerline of the sutures **14**, **22** and the first ends **16**, **24**. The stuffing **44** may be disposed adjacent to the spliced segment **254**, **256** between the shuttle suture **22** and the repair suture **14** that forms the locking feature **32** depending on the configuration of the suture assembly **10**D, **10**E.

[0169] The shuttle suture **22** at or proximate to the looped end **26** may be spliced into the repair suture **14** (**324**). The second spliced segment **254** between the repair suture **14** and the shuttle suture **22** may be formed (**326**). The shuttle loop **30**D may be included in the spliced segment **254**. The second end **26**, including the end shuttle loop **30**D, of the shuttle suture **22** may be substantially or fully encapsulated in the repair suture **14**. The second end **18** of the repair suture **14** may extend beyond the second end **26** of the shuttle suture **22**. Alternatively, the shuttle loop **30**D may remain a loose tail end, and the spliced segment **254** may be spaced from the shuttle loop **30**D. In the suture assembly **10**E, the second spliced segment **254** may form the locking feature **32**.

[0170] In certain aspects, such as for the suture assembly **10**D, the mid-section **200** of the shuttle suture **22** may be spliced into the mid-region **258** of the repair suture **14** (**328**). In such aspects, the third spliced segment **256** may be formed between the shuttle suture **22** and the repair suture **14** (**330**). The shuttle suture **22** and the repair suture **14** may be spliced together at mid-section **200** and the mid-region **258** to form the locking feature **32**. The third spliced segment **256** may be formed proximate to the stuffing **44** or stuffing suture **44**. The steps **328**, **330** may be omitted for the suture assembly **10**E with the two spliced segments **252**, **254** with the shuttle suture **22**. However, any number of spliced segments may be utilized without departing from the teachings herein. Further, while the locking feature **32** is described as the spliced segments **254**, **256** to pull the repair suture **14** through itself, the grommet **80** (FIG. **10**) may be used without departing from the teachings herein.

[0171] Excess length or “slack” of the shuttle suture **22** may be reduced or removed by applying a pulling force to the shuttle suture **22** (**332**). In configurations with the free looped end **26**, the pulling force may be applied to the looped end **26**. The pulling force may allow the shuttle suture **22** to move relative to and/or through the repair suture **14** at the spliced segments **252**, **254**, **256**.

[0172] The ends **16**, **18** of the repair suture **14** may be sealed or formed (**334**). The limbs **264**, **266**, **268**, **276**, **278** at each end **16**, **18** may be joined or combined together, such as via adhesive(s). The suture assembly **10** may be completed with the shuttle suture **22** spliced into the repair suture **14** at two segments **252**, **254** or three segments **252**, **254**, **256** and the stuffing suture **44** spliced into the repair suture **14** at the spliced region **262**. One or both needles **20**, **28** may be coupled with the suture assembly **10** (**336**). The steps **312**-**336** of the method **310** may be modified or conducted in different orders, with steps omitted, repeated, performed concurrently, etc. without departing from the teachings herein. For example, the spliced segments **252**, **254**, **256** and spliced region **262** may be formed in any order without departing from the teachings herein.

[0173] The pre-combined suture assembly **10**, including the exemplary suture assemblies **10**D, **10**E, may be advantageous for stitching or suturing multiple tissue bodies T**1**, T**2** together and for implant attachment procedures. The pre-combined suture assembly **10** may maximize the efficiency of the procedure and reduce strands of sutures **14**, **22** that are initially inserted through the tissue T, which may reduce resistance in the insertion process. The pre-combined suture assembly **10** may be utilized with two needles **20**, **28** for inserting both ends of the suture assembly **10** through the tissue T and/or the implant, similar to the process described in FIGS. **4-8**. Two needles **20**, **28** may be advantageous when the pledget **12** is utilized to position the pledget **12** against the underside of the tissue T. Additionally or alternatively, the pre-combined suture assembly **10** may be utilized one needle **20**. The single needle **20** configuration may be advantageous for inserting a single end of the suture assembly **10** through tissue T, which may be advantageous when the suture assembly **10** omits the pledget **12**.

[0174] Referring to FIGS. **25-37**, when suturing tissue bodies T**1**, T**2**, the tissue bodies T**1**, T**2** may be two separate tissues or may be portions of the same tissue T on opposing sides of an incision, cut, or other spacing. Moreover, the tissue bodies T**1**, T**2** may be side-by-side to one another (see FIGS. **25-31**) or vertically overlap one another (see FIGS. **32-37**). By way of example, the suture assembly **10**E is illustrated in a process of connecting two tissue bodies T**1**, T**2**.

[0175] Referring to FIGS. **25-31**, the suture assembly **10** may be utilized to attach or connect two laterally adjacent tissue bodies T**1**, T**2**. The tissue bodies T**1**, T**2** may be arranged side-by-side with the incision or cut therebetween. The lead end **16** of the suture assembly **10**, which may be or include the lead end **16** of the repair suture **14**, may be inserted through the first tissue body T**1** in a first or outside-to-inside direction. The lead end **16** may be tapered and may carry the first end **24** of the shuttle suture **22** encapsulated therein. The ends **16**, **24** may be carried via the needle **20** or the insertion device. The encapsulated end **24** of the shuttle suture **22** may be advantageous for carrying two sutures **14**, **22** through the tissue T at a single time and as a single limb. Further, the encapsulated end **24** may reduce friction or resistance as the suture assembly **10** is carried through the tissue T. For example, the lead end **16** of the repair suture **14** may be thinner or smaller than where the end **24** of the shuttle suture **22** is encapsulated, which may result in a thinner or smaller portion of the suture assembly **10** initially being inserted through the tissue T to prepare for the thicker portion of the suture assembly **10**.

[0176] Referring to FIGS. **25** and **26**, the lead end **16** of the suture assembly **10** may be carried or inserted through the second tissue body T**2**. Generally, the suture assembly **10** may be moved under the tissue T and across the cut or incision between the tissue bodies T**1**, T**2**. The suture assembly **10** may be inserted in a second opposing or inside-to-outside direction through the second tissue body T**2** for the ends **16**, **24** of the repair suture **14** and the shuttle suture **22** to be arranged on an outer side of the tissue T.

[0177] When utilizing the suture assembly **10** for the tissue bodies T**1**, T**2**, the suture assembly **10** is inserted or carried through the tissue bodies T**1**, T**2** in opposing directions (e.g., in the first direction and also the second direction). The configuration of inserting a single end of the suture assemblies **10** through the two tissue bodies T**1**, T**2** may be advantageous for minimizing the size of the suture assembly **10** that is carried through the tissue T. For example, using the suture assembly **10**E for illustrative purposes, the end of the suture assembly **10** with the shuttle end loop **30**D may be larger or thicker, which may be more difficult to carry through the tissue T or result in more resistance/friction. However, each end of the suture assembly **10** may be carried from an inner side to an outer side without departing from the teachings herein, such as when both ends of the suture assembly **10** are coupled with needles **20**, **28**.

[0178] Referring now to FIGS. **27** and **28**, the suture assembly **10** may continue to be pulled through the tissue bodies T**1**, T**2** by applying the pulling force or tension to the lead ends **16**, **24** of the sutures **14**, **22**. The pulling force may be applied until the locking feature **32** and the thickened portion formed by the stuffing suture **44** (e.g., the spliced region **262** and/or the mid-region **258**) are disposed between the tissue bodies T**1**, T**2**. The locking feature **32** may be centrally located between the tissue bodies T**1**, T**2**. The spliced region **262** with the stuffing suture **44** may be arranged below the tissue T and between the two tissue bodies T**1**, T**2**. As illustrated, the spliced region **262** may be closer to one of the tissue bodies T**2**. Accordingly, the spliced region **262** may be generally arranged at the cut or incision prior to the shuttling process.

[0179] The lead end **16** of the repair suture **14** may be separated from the end **24** of the shuttle suture **22**. The suture assembly **10** may be cut to remove the needle **20**, which may also separate the sutures **14**, **22**. The end **24** of the shuttle suture **22** may be moved from within the spliced segment **254** to be a loose tail end **24**. Alternatively, one or more of the limbs **264**, **266**, **268** may be separated from the combined end **16** of the repair suture **14** to provide access to the shuttle suture **22**. The lead end **16** of the repair suture **14** may be independently movable relative to the end **24** of the shuttle suture **22**.

[0180] Referring now to FIGS. **28** and **29**, the lead or moving end **16** of the repair suture **14** may be threaded through the shuttle end loop **30**D, extending across the cut or incision. The first end **24** of the shuttle suture **22** may remain proximate to the second tissue body T**2**. The repair suture **14** may include one or more fold indicator(s) **56**, which may provide visual feedback on where to fold the suture **14**. The lead end **16** of the repair suture **14** may be inserted through the shuttle end loop **30**D and folded back at the fold indicator **56**. The repair suture **14** may generally fold back on itself to extend along itself to form the folded end **16** around the shuttle end loop **30**D. This folded arrangement around the shuttle end loop **30**D may allow the repair suture **14** to remain engaged with the shuttle end loop **30**D as the pulling force is applied to the tensioning end **24** of the shuttle suture **22**. The lead end **16** of the repair suture **14** may be tapered, which may be advantageous for forming a smaller folded end **16** for guiding the folded end **16** through the tissue T and the locking feature **32**.

[0181] The pulling force can be applied to the tensioning end **24** of the shuttle suture **22**, which may draw the shuttle end loop **30**D and the engaged folded end **16** of the repair suture **14** toward and through the first tissue body T**1**. A holding force may be applied to the opposing static end **18** of the repair suture **14** to move a portion of the repair suture **14** during the shuttling process while retaining engagement between the repair suture **14** and the tissue bodies T**1**, T**2**. During the shuttling process, the repair suture **14** may be looped over the spacing between the tissue bodies T**1**, T**2** and drawn through the first tissue body T**1** from the outer side to the inner side and toward the locking feature **32**.

[0182] The locking feature **32**, which may be the second spliced segment **254**, may be the location where the shuttle suture **22** is interlaced, woven, or otherwise extended through the repair suture **14**. The pulling force at the tensioning end **24** of the shuttle suture **22** may be configured to draw the shuttle suture **22** and the engaged folded end **16** through the spliced segment **254**. With the tension or pulling force being applied to the end **24** of the shuttle suture **22**, the shuttle suture **22** may be drawn through the spliced segment **254** or other locking feature **32**. As the shuttle end loop **30**D is drawn through spliced segment **254**, the folded end **16** of the repair suture **14** may also be drawn through the spliced segment **254** or locking feature **32**. This movement may pull the shuttle suture **22** out of the repair suture **14** and the tissue T and the repair suture **14** through itself. The tapered nature of the end **16** of the repair suture **14** may be advantageous for pulling the repair suture **14** through itself with less resistance.

[0183] The repair suture **14** may include the thickened portion with the stuffing suture **44** proximate to the locking feature **32** where the repair suture **14** is pulled through itself. The repair suture **14** being drawn through the spliced segment **254** and tensioning of the suture **14** may cause the stuffing suture **44** to compress, wrinkle, or bunch within the mid-region **258** of the repair suture **14**. The bunching of the thickened portion may form a locking trap (e.g., the locking feature **32**) that assists with locking the repair suture **14** in the looped or stitch configuration. The resistance caused by the thicker stuffed portion may lock the repair suture **14** under tension. The locking trap may allow re-tensioning or further tightening while reducing or preventing loosening of the repair suture **14**.

[0184] Referring now to FIGS. **28-31**, the pulling force may continue to draw the shuttle end loop **30**D and the folded end **16** through the second tissue body T**2** from the inner side to the outer side, forming a full loop with the repair suture **14** across and between the tissue bodies T**1**, T**2**. The shuttle suture **22** may be fully removed from the tissue T upon completion of the shuttling process. The pulling force can be applied to the lead end **16** of the repair suture **14** to tension or tighten the repair suture **14**, drawing or compressing the tissue bodies T**1**, T**2** together. The pulling force can be applied until the tissue bodies T**1**, T**2** abut or overlap one another, closing the incision between the tissue bodies T**1**, T**2**. In other words, the pulling force can be applied until sufficient compression occurs between the tissue bodies T**1**, T**2**. The knotless locking feature **32** may retain the repair suture **14** under tension and, consequently, the compression between the tissue bodies T**1**, T**2**. The ends **16**, **18** of the repair suture **14** may then be cut proximate to the tissue T to reduce excess suture **14** length in the patient anatomy.

[0185] Referring to FIGS. **32-37**, similarly, the suture assembly **10** may also be used for tissue bodies T**1**, T**2** that are stacked over one another. In such examples, the tissue bodies T**1**, T**2** may be vertically stacked or overlapped with the cut, incision, or spacing separating two layers of tissue bodies T**1**, T**2**. The lead end of the suture assembly **10**, including the lead ends **16**, **24** of the sutures **14**, **22**, may be inserted through the first outer tissue body T**1**, across the spacing, and through the second inner tissue body T**2**. Generally, the two inward insertion locations (illustrated on the left side in FIGS. **32** and **33**) may be aligned with one another to allow movement of the tissue bodies T**1**, T**2** toward one another while reducing side-to-side movement between the tissue bodies T**1**, T**2**, which may increase stress on the tissue T.

[0186] Referring still to FIGS. **32-35**, the suture assembly **10** may extend along an underside of the inner tissue body T**2**. The lead end **16** may be inserted or carried through the inner tissue body T**2**, across the spacing, and through the outer tissue body T**1**. The two outward through locations (illustrated on the right side in FIGS. **32** and **33**) may generally be parallel with one another to provide a compression between the tissue bodies T**1**, T**2** with reduced side-to-side movement. The suture assembly **10** may extend parallel to itself as it extends through the tissue bodies T**1**, T**2**, which may reduce bunching or lateral movement of the tissue bodies T**1**, T**2**.

[0187] The pulling force may be applied to the lead end of the suture assembly **10** until the locking feature **32** and the spliced region **262** with the stuffing suture **44** are disposed below the inner tissue body T**2**. The thickened portion may be advantageous under the tissue T for distributing the force applied by the repair suture **14** on the tissue T. The ends **16**, **18**, **24**, **26** of the shuttle suture **22** and the repair suture **14** may be adjusted to allow independent movement of the repair suture **14** relative to the shuttle suture **22**. For example, the needle **20** may be cut from the suture assembly **10**, allowing separation of the sutures **14**, **22**.

[0188] Referring to FIGS. **35-37**, the shuttle suture **22** may be utilized for the shuttling process. The lead end **16** of the repair suture **14** may be moved across the outer tissue body T**2**, inserted through the shuttle end loop **30**D, and folded back on itself. The pulling force may be applied to the tensioning end **24** of the shuttle suture **22** to pull the shuttle end loop **30**D and, consequently, the folded end **16** of the repair suture **14**. The folded end **16** may be drawn by the shuttle suture **22** in a path including: through the outer tissue body T**1**, across the spacing, through the inner tissue body T**2**, along the underside of the inner tissue body T**1** and through the locking feature **32**/spliced segment **254**, through the inner tissue body T**2**, across the spacing, and through the outer tissue body T**1**. This path may form the looped or stitch configuration to compress the tissue bodies T**1**, T**2** together and remove the shuttle suture **22** from the tissue T.

[0189] The repair suture **14** may be drawn through itself at the spliced segment **254** and cause the stuffing suture **44** to bunch, forming the locking trap for locking the repair suture **14** under tension in the stitch configuration. The pulling force may be applied to the lead end **16** of the repair suture **14** to compress the tissue bodies T**1**, T**2** together. The knotless locking feature **32** may retain the repair suture **14** under tension and the compression between the tissue bodies T**1**, T**2**. Upon sufficient compression between the tissue bodies T**1**, T**2**, the ends **16**, **18** of the repair suture **14** may be trimmed or cut proximate to the tissue T.

[0190] The suture assembly **10** may be a knotless orthopedic suture assembly **10** for connecting multiple tissue bodies T**1**, T**2**, which may be separate tissues T or portions of the same tissue T. The suture assembly **10** may utilize the knotless locking feature **32**, which can hold the repair suture **14** under tension without the use of more traditional knots and crimps. This may reduce the profile of the suture assembly **10** that remains in the patient anatomy and may reduce implants in the patient. The suture assembly **10** may combine the two sutures **14**, **22**, allowing for increased efficiency in carrying the sutures **14**, **22** through the tissue T and minimizing resistance caused by the suture assembly **10**. Though discussed primarily in reference to connecting tissue bodies T**1**, T**2**, it shall be understood that knotless suture assembly **10** may be applied in a variety of applications and procedures.

[0191] Referring to FIG. **38**, as well as FIGS. **25-37**, the suture assembly **10** may be utilized for orthopedic procedures and methods (**360**) to connect tissue bodies T**1**, T**2** at an incision, cut, or other spacing. For suturing multiple tissue bodies T**1**, T**2**, the suture assembly **10** can be provided (**362**), which may be formed using the method of manufacturing (**310**) described herein with respect to FIG. **24** or another manufacturing method. The suture assembly **10** may be preassembled or preformed with the repair suture **14** and the shuttle suture **22** spliced together. The lead end of the suture assembly **10** may be carried or inserted through the tissue T in the first or outward-to-inward direction (**364**). Typically, the lead end may be the combined first ends **16**, **24** of the sutures **14**, **22**. The suture assembly **10** may be carried through the first tissue body T**1** when the tissue bodies T**1**, T**2** are laterally adjacent to one another or both tissue bodies T**1**, T**2** when the tissue bodies T**1**, T**2** overlap.

[0192] In certain aspects, the suture assembly **10** may be inserted through the pledget **12** at the underside of the tissue T. This may allow the suture assembly **10** to be inserted in a single direction via one needle **20** and include the pledget **12**. The needle **20** may be used to insert or carry the suture assembly **10** through the pledget **12**. The pledget **12** may assist with distributing pressure from the repair suture **14** and may assist in closing the opening between the tissue bodies T**1**, T**2**.

[0193] The lead end of the suture assembly **10** may be carried or inserted through the tissue T in the second opposing or inward-to-outward direction (**366**). This may be the second tissue body T**2** when the tissue bodies T**1**, T**2** are laterally adjacent to one another or both tissue bodies T**1**, T**2** when the tissue bodies T**1**, T**2** overlap. A pulling force may be applied to the lead end of the suture assembly **10** to position the locking feature **32** and the spliced region **262** with the stuffing suture **44** below the tissue T (**368**). The insertion or lead ends **16**, **24** of the repair suture **14** and the shuttle suture **22** may be separated or otherwise adjusted to allow independent movement of the sutures **14**, **22** relative to one another (**370**). The lead end **16** of the repair suture **14** may be moved across the tissue T and inserted through the shuttle end loop **30**D (**372**). The repair suture **14** may be folded back on itself (**374**). The pulling force may be applied to the first end **24** of the shuttle suture **22** to move the repair suture **14** along a shuttling path (**376**).

[0194] The folded end **16** of the repair suture **14** may be guided through the tissue T (**378**) (e.g., one or both of the tissue bodies T**1**, T**2**) in the outward-to-inward direction and through the spliced segment **252** (**380**), engaging the locking feature **32** and causing the thickened portion to bunch and lock the repair suture **14** (**382**). The folded end **16** of the repair suture **14** may be guided through the tissue T (e.g., one or both of the tissue bodies T**1**, T**2**) in the inward-to-outward direction, and the shuttle suture **22** may be withdrawn or removed from the tissue T (**384**). The pulling force may be applied to the lead end **16** of the repair suture **14** (**386**) and the tissue bodies T**1**, T**2** can be compressed together (**388**). During the shuttling and tensioning processes, the second end **18** of the repair suture **14** may remain static. Alternatively, the pulling force may also be applied to the second end **18** to assist with tensioning the repair suture **14** and compressing the tissue bodies T**1**, T**2** together. The repair suture **14** may be re-tensioned (**390**) and the ends **16**, **18** trimmed or cut (**392**). The steps **362-392** of the method **360** may be modified or conducted in different orders, with steps omitted, repeated, performed concurrently, etc. without departing from the teachings herein.

[0195] The suture assembly **10** may be inserted through the tissue T or tissue bodies T**1**, T**2** multiple times prior to the shuttling process without departing from the teachings herein. This may allow the formation of multiple "stitches" formed from the same suture assembly **10**, which may be advantageous for providing more connection points between two tissue bodies T**1**, T**2**. More connection points may assist with providing more even compression between the tissue bodies T**1**, T**2** and/or providing more connections along a longer incision or opening between tissue bodies T**1**, T**2**. Additionally or alternatively, multiple suture pairs **40**, each formed from an integrated suture assembly **10**, may be utilized along a single incision or cut between tissue bodies T**1**, T**2**. The ability to re-tension the repair sutures **14** may be advantageous when utilizing multiple suture pairs **40** to distribute pressure and provide better alignment between the tissue bodies T**1**, T**2**.

[0196] While described with a single insertion end, both ends of the suture assembly **10** may be inserted through the tissue T. In such examples, the first needle **20** carrying the first ends **16**, **24** can be inserted through the first tissue body T**1**, and the second needle **28** carrying the second ends **18**, **26** can be inserted through the second tissue body T**2**. The locking feature **32** may be arranged below and/or between the tissue bodies T**1**, T**2**. When using two needles **20**, **26**, the pledget **12** may also be used. The needles **20**, **28** may be removed from the suture assembly **10** and the shuttling process may be performed to compress the tissue bodies T**1**, T**2** together. This process may be substantially similar to the steps **104-124** of the method **100** described with reference to FIG. **11**, as well as the methods **150**, **220** in FIGS. **14** and **17**, respectively.

[0197] The suture assembly **10** may be advantageous for inserting a single combined assembly **10** through the tissue T to then be secured via the knotless configuration. The suture assembly **10** may be advantageous for orthopedic procedures, joining the multiple tissue bodies T**1**, T**2** together and maintaining compression therebetween. The knotless configuration may allow the repair suture **14** to be re-tensioned, continuing to apply sufficient compression between the tissue bodies T**1**, T**2**. The suture assembly **10** may also have a smaller overall profile when omitting the pledget **12** and may have a smaller profile for being drawn through the tissue T combining the sutures **14**, **22** together. The suture assembly **10** may also be used for valve replacement procedures, where the suture assembly **10** may attach the tissue T with the valve in a similar manner as the two tissue bodies T**1**, T**2** (e.g., in a single direction with one needle **20**) or similar to the processes described above with the suture assemblies **10**A-**10**C.

[0198] Referring again to FIGS. **1-38**, the suture assemblies **10** described herein may be re-tensioned during the valve replacement procedure, implant attachment procedure, and tissue connecting procedures. This may allow for an increased seal, more accurate alignment, such as between the valve V and the tissue T, and/or increased compression between the tissue bodies T**1**, T**2**. Using a valve replacement procedure as an example, conventional sutures are generally tensioned a single time without the ability to be re-tensioned, which can result in limitations in adjusting the valve V and, consequently, improper alignment between the valve V and the tissue T. Often, during the valve replacement procedure, the repair sutures **14** are initially provided at the commissures C where the leaflets meet. For an aortic valve replacement, there are generally three commissures C (see FIG. **2**). With more conventional sutures, the sutures are generally tightened in sequence at the commissures C and, once tightened, the sutures can be knotted or a crimp is crimped to lock or hold the sutures. Once the suture is knotted or the crimp is crimped, the sutures are generally fixed and may not be re-tensioned or adjusted. If the valve V shifted during the procedure, which can be common based on tension being applied to the conventional sutures in sequences, the valve V is generally fixed in the slightly displaced position.

[0199] The re-tensioning that can be accomplished via the suture assemblies **10** disclosed herein may provide for better alignment of the replacement valve V. For example, using the suture assembly **10**, the repair sutures **14** may initially be tensioned at the commissures C to attach the replacement valve V to the tissue T. During this process, the replacement valve V may become displaced due to the temporarily uneven tension of the repair sutures **14**, which can be more common in mechanical valves V. Each of the suture assemblies **10** can then be re-tensioned multiple times during the valve replacement procedure to assist with balancing the repair or fixation of the replacement valve V and positioning of the replacement valve V. Accordingly, the ability to re-tension the repair suture **14** may allow for better alignment of the valve V, which can also result in a better fluid seal between the valve V and the tissue T. The suture assemblies **10** can also be re-tensioned for other implant procedures and for connecting tissue bodies T**1**, T**2**.
[0200] Further, in various aspects, the suture assemblies **10** may also be utilized with implant clips. In such configurations, the implant clips may engage the looped repair suture **14** formed from the shuttling process before the compression and tensioning in the stitch configuration. The implant clips may engage adjacent repair sutures **14**, which may provide a greater surface area for compression between the tissue bodies T**1**, T**2** and/or between the valve V and the tissue T.
[0201] Referring still to FIGS. **1-38**, in certain aspects, the suture assembly **10** may also include smaller/thinner sutures **14**, **22**, which may reduce the footprint of the suture assembly **10**. One or both of the sutures **14**, **22** may be 0, 2-0, 4-0, or 5-0 sutures depending on the configuration of the suture assembly **10**. This sizing may provide a balance between being sufficiently narrow for the sutures **14**, **22** to be threaded through various features, including the locking feature **32**, and having a sufficient thickness to form the spliced section **70** and/or spliced segments **252**, **254**, **256** of the repair suture **14**. This sizing can also provide sufficient size for the bifurcated section of the shuttle link **30**A of the shuttle suture **22**, the shuttle receiving area **30**B, the folded shuttle loop **30**C, and/or the end shuttle loop **30**D when used. The sutures **14**, **22** may have a variety of configurations, such as tape-like, flat, tubular, tubular that compresses flat, etc. to balance between size and function. The sutures **14**, **22** may also be manufactured to promote multiple spliced segments **252**, **254**, **256** between the sutures **14**, **22** to form a combined or integrated assembly **10**.
[0202] Additionally, different portions of the sutures **14**, **22** may have different configurations. For example, the repair suture **14** may have a tape-like portion that may become pledget-like, which can include the spliced section **70**, the spliced segments **252**, **254**, **256**, and/or the spliced region **262**. The pledget-like section of the repair suture **14** may be coupled with the pledget **12** and/or replace the pledget **12**. The pledget-like section may be advantageous in the suture assemblies **10**D, **10**E to better distribute pressure from the repair suture **14** while providing an integrated assembly **10** formed with a single insertion end.
[0203] The suture assemblies **10** can also include the pledget **12**, which may also have a variety of configurations, including the flat pledget **12**A, the folded pledget **12**B, and the annular pledget **12**C. For example, the pledget **12** may include a braided jacket to be built into the round suture, similar to a fiber tape. This may further reduce the size of the overall assembly **10** and number of components utilized in the replacement procedure.
[0204] Additionally, the sutures **14**, **22** may have increased strength compared to conventional sutures. One or both of the repair suture **14** and the shuttle suture **22** may be constructed of polyethylene. The polyethylene construction may be more inert and stronger than a conventional polyester suture. The increased strength may be advantageous for applying the pulling force to the sutures **14**, **22** during the shuttling process where conventional polyester sutures may have an increased risk of shredding or breaking down during this process. The use of polyethylene may provide the strength profile for the shuttling process.
[0205] Referring still to FIGS. **1-38**, one or both of the repair sutures **14** and the shuttle sutures **22** may be manufactured to have different visual indicators or differentiators to guide the use of the suture assembly **10** during the valve replacement procedure. These visual indicators include the cut

indicators **50-54**, the fold indicator(s) **56**, and/or the insertion indicator(s) **132**. Additionally, the visual differentiation may assist with distinguishing the shuttle suture **22** from the repair suture **14**, as well as the location on the shuttle suture **22** to apply the pulling force. For example, the two sutures **14**, **22** may be manufactured in different colors or with different patterns along the length of the sutures **14**, **22**. Additionally, for the suture assemblies **10**A, **10**B, the first end **24** of the shuttle suture **22** may have a different pattern than the second end **26**, such as black stripes on one side of the shuttle suture **22**, which may be advantageous for indicating the pulling location for the shuttle suture **22**. The differentiation may also be helpful when separating spliced sutures **14**, **22** for the shuttling process.

[0206] The visual indicators and differentiators may assist with providing markers for guiding the procedure, such as the cut indicators **50-54**, the fold indicator(s) **56**, and/or the insertion indicator(s) **132**. The differentiation between the two sutures **14**, **22** and identifying the pulling location may be advantageous when multiple suture pairs **40** are arranged around a single annulus A or incision between tissue bodies T**1**, T**2**. This may assist in properly and more efficiently identifying the different sutures **14**, **22**, particularly if two sutures **14**, **22** from different suture pairs **40** become crossed during the replacement procedure.

[0207] Referring still to FIGS. **1-38**, the shuttle sutures **22** and the repair sutures **14** may be manufactured to have different widths and different components. The shuttle sutures **22** may be woven as a single limb, bifurcated into two limbs to form the shuttle link **30**A, and then again woven as a single limb. This manufacturing process may allow the shuttle link **30**A to be formed without significantly increasing the width of the shuttle suture **22** at the shuttle link **30**A. Alternatively, the shuttle sutures **22** may be folded to form two limbs that define the folded shuttle loop **30**C or the end shuttle loop **30**D. Further, the repair suture **14** may include one or more of multiple limbs **264**, **266**, **268**, **276**, **278**, transition sections, **270**, **272**, **282**, spliced segments **252**, **254**, **256**, and the spliced region **262**.

[0208] In additional non-limiting examples, both the sutures **14**, **22** may have one limb along a substantial portion, or an entirety, of their lengths. When folding the shuttle suture **22** to form the shuttle loops **300**, **30**D or using the single limb with the shuttle receiving area **30**B, the width/thickness of the shuttle suture **22** may be smaller than when using the bifurcated shuttle link **30**A. Additionally, during a weaving, braiding, or interlacing process for the repair suture **14**, a number of strands in the repair suture **14** may be reduced to form the narrower and thinner first end **16** and/or second end **18**. The different visual indicators may also be added or incorporated into the sutures **14**, **22** during one or more stages of the manufacturing process.

[0209] As described herein, the first and second needles **20**, **28** may be utilized to carry portions of the sutures **14**, **22** through the tissue T and/or the valve V before being removed from the suture assembly **10**. It is contemplated that certain devices may be utilized in lieu of the needles **20**, **28** for carrying the sutures **14**, **22** through the tissue T and/or the implant without departing from the teachings herein. In such examples, the device may function similarly to the needles **20**, **28** for engaging the sutures **14**, **22**, carrying the sutures **14**, **22**, and then disengaging from the sutures **14**, **22** for the shuttling process. In other words, the needles **20**, **28** may not be included in the suture assembly **10**, rather the device is utilized with the sutures **14**, **22**.

[0210] For example, a suture passer device may be utilized with the suture assembly **10**. The suture passer device may include a distal jaw with at least one movable jaw member and a movable needle. The jaw may be utilized for holding and moving the tissue. The jaw may also support the needle, which can be selectively engaged with the suture **14**, **22** to be passed. In certain aspects, a user may load one or both of the sutures **14**, **22** onto the needle of the suture passer. Additionally or alternatively, the suture passer may be configured to automatically re-load one or both of the sutures **14**, **22** onto the needle. The suture passer may be used for carrying the sutures **14**, **22** through the tissue T and/or the valve V without using an attached needle **20**, **28**. In other words, the needle can remain as part of the suture passer, omitting the process of removing the needles **20**, **28** from the suture assembly **10**.

[0211] In an additional non-limiting example, an implant device can be used with the suture assembly **10** for placing the suture assembly **10** in the tissue T. The implant device may carry or house one or both sutures **14**, **22** and/or the pledget **12**. The suture assembly **10** may be housed in a sheath of the implant device. A needle tip of the implant device can be inserted through the tissue T and/or valve V and the user can deploy a portion of the suture assembly **10**, such one section as the sutures **14**, **22**. This first section of the sutures **14**, **22** may be the portions coupled with the first needle **20** in configurations utilizing the attached needles **20**, **28**. The user can then adjust the implant device to load a second section of the suture assembly **10** to be implanted. The second section of the sutures **14**, **22** may be the portions coupled with the second needle **28** in configurations utilizing the attached needles **20**, **28**. The implant device can be removed from the tissue T and then re-inserted through the tissue T at a second location to deploy or carry the second section of the sutures **14**, **22** through the tissue T at the second location to form the stitch arrangement.

[0212] The suture assembly **10** disclosed herein generally includes the attached needles **20**, **28** for carrying the sutures **14**, **22** through the tissue T and/or the valve V for forming the knotless stitch configuration. The suture assembly **10** may also utilize a device for passing the sutures **14**, **22**, such as the suture passer device or implant device, rather than one or both of the attached needles **20**, **28**. The suture assembly **10** may be configured substantially similarly with and without the attached needles **20**, **28** for forming the stitch configuration with the shuttling process to provide a knotless stitch that can be tensioned and re-tensioned.

[0213] Referring still to FIGS. **1-38**, the suture assemblies **10** disclosed herein and use of the assemblies **10** may provide for a variety of advantages. For example, the suture assemblies **10** may include two needles **20**, **28** for each suture pair **40**, which can allow the pledget **12** to be disposed under the tissue T. Also, the two needles **20**, **28** may allow both of the sutures **14**, **22** to be directed through the tissue T and the replacement valve V, which can increase engagement and compression between the tissue T and the replacement valve V to provide a better fluid-tight or blood-tight seal for high-pressure blood flow. The suture assemblies **10** may also be used with a single needle **20** for a single end being inserted through the tissue T, which may be advantageous for orthopedic approaches, such as connecting issue bodies T**1**, T**2**. The suture assembly **10** may be used with the pledget **12**, which may provide pressure distribution and reinforcement for the tissue T during the tensioning and re-tensioning processes. The suture assembly **10** may also be used without the pledget **12** for single-direction insertion or other procedures.

[0214] Additionally, one or both of the sutures **14**, **22** may be constructed of polyethylene to increase the strength profile of the sutures **14**, **22** for the shuttling process compared to more conventional polyester sutures. Further, the shuttle suture **22** may be bifurcated to form the shuttle link **30**A, may have the shuttle receiving area **30**B, or may be folded to form the folded shuttle loops **30**C, **30**D. The repair suture **14** may have the narrower end(s) **16**, **18** to assist with the sutures **14**, **22** being drawn through the locking feature **32**, such as through the spliced section **70** or spliced sections **254**, **256** of the repair suture **14**, with reduced or minimal resistance. Moreover, the suture assemblies **10** disclosed herein may reduce or eliminate metal components, such as more conventional crimps. In this regard, the suture assemblies **10** may also reduce or eliminate components that remain on the outer side of the replacement valve V (opposite the side that engages the tissue T) or the tissue T, where traditional knots and crimps can remain. The suture assemblies **10** may each be a knotless assembly **10**, which may be more convenient and efficient when operating in smaller areas or where multiple suture assemblies **10** are utilized.

[0215] Additionally, the pre-connected sutures **14**, **22** may allow the surgeon to insert a single assembly **10** through the tissue T and secure the tissue bodies T**1**, T**2** or the tissue T and the implant (e.g., the valve V) with a knotless suture configuration. The suture assembly **10** may also have a smaller overall profile when omitting the pledget **12** and may have a smaller profile for being drawn through the tissue T by being integrated or combined. The suture assembly **10** may include one end **24** of the shuttle suture **22** encapsulated in the repair suture **14** or both ends **24**, **26** encapsulated in the repair suture **14**. Different methods may be used with the suture assembly **10** depending on the configuration, such as one needle **20** with one end being inserted through tissue T, two needles **20**, **28** with both ends being inserted through tissue T, or the use of an insertion device. The suture assemblies **10** may be utilized for attaching implants and for orthopedic procedures, such as attaching tissue bodies T**1**, T**2** together. Additional benefits or advantages would be realized and/or achieved.

[0216] The device disclosed herein is further summarized in the following paragraphs and is further characterized by combinations of any and all various aspects described herein.

[0217] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget. A repair suture may have a proximal end fixedly coupled to the pledget and a distal end coupled to a first needle. A shuttle suture may extend through the pledget. The shuttle suture may have a first end coupled to the first needle and a second end coupled to a second needle. The shuttle suture may include a shuttle link proximate to the second needle. A locking feature may be coupled to the pledget. The locking feature may be configured to retain the repair suture under tension when the repair suture is threaded through the locking feature.

[0218] According to an aspect of the present disclosure, a shuttle suture may include a bifurcated section that forms a shuttle link.

[0219] According to an aspect of the present disclosure, a locking feature may be a spliced section of a repair suture disposed within a pledget, and a shuttle suture may be configured to extend through the spliced section.

[0220] According to an aspect of the present disclosure, a locking feature may be a grommet including outer barbs to engage a pledget.

[0221] According to an aspect of the present disclosure, a distal end of a repair suture may taper to a lesser width compared to a proximal end.

[0222] According to an aspect of the present disclosure, at least one of a shuttle suture and a repair suture may be constructed of polyethylene.

[0223] According to an aspect of the present disclosure, a shuttle suture may be visually differentiated from a repair suture by at least one of a color and a pattern.

[0224] According to an aspect of the present disclosure, a first end of a shuttle suture may be visually differentiated from a second end of the shuttle suture by at least one of a color and a pattern.

[0225] According to an aspect of the present disclosure, a pledget may be folded to form a first layer and a second layer, and a repair suture and a shuttle suture may extend through a middle of the first layer and the second layer.

[0226] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget defining a ring shape and multiple suture pairs coupled to the pledget. Each suture pair may include a repair suture having a proximal end that may be fixedly coupled to the pledget and a distal end that may be coupled to a first needle, as well as a shuttle suture that may have a first end coupled to the first needle and a second end that may be coupled to a second needle. The shuttle suture may extend through the pledget proximate to the repair suture. The shuttle suture may include a shuttle link proximate to the second needle to receive the distal end of the repair suture.

[0227] According to an aspect of the present disclosure, a method of replacing a cardiovascular valve may include one or more of: inserting a first needle through tissue and a replacement valve, wherein the first needle is coupled to a distal end of a repair suture and a first end of a shuttle suture; inserting a second needle through the tissue and the replacement valve to position a pledget against an inner surface of the tissue, wherein the second needle is coupled to a second end of the shuttle suture; removing the first and second needles; inserting the distal end of the repair suture through a shuttle link of the shuttle suture; applying a pulling force to the first end of the shuttle suture to pull the second end of the shuttle suture and the distal end of the repair suture through the pledget and a locking feature-coupled with the pledget; and tensioning and locking the repair suture.

[0228] According to an aspect of the present disclosure, a method may include a step of tensioning and locking a repair suture, which may include threading the repair suture through a spliced section of the repair suture to lock the repair suture.

[0229] According to an aspect of the present disclosure, a method may include re-tensioning a repair suture.

[0230] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget, a repair suture, and a shuttle suture. The repair suture may have a proximal end fixedly coupled to the pledget and a distal end coupled to a first needle. The repair suture may include a locking feature. The shuttle suture may have a first end and a second end each coupled to the first needle. The shuttle suture may be folded and may extend through the locking feature to form a shuttle loop. The shuttle loop may be operably coupled to a second needle via a connecting suture. The shuttle loop may be configured to guide the repair suture through the locking feature to retain the repair suture under tension when the repair suture is threaded through the locking feature.

[0231] According to an aspect of the present disclosure, a locking feature may be a spliced section of a repair suture.

[0232] According to an aspect of the present disclosure, a shuttle suture may be a monofilament suture.

[0233] According to an aspect of the present disclosure, a method of replacing a cardiovascular valve may include one or more of: inserting a first needle through tissue and a replacement valve, where the first needle is coupled to a distal end of a repair suture and a first and second ends of a shuttle suture; inserting a second needle through the tissue and the replacement valve to position a pledget against an inner surface of the tissue, where the second needle is coupled to a shuttle loop of the shuttle suture; removing the first and second needles; inserting the distal end of the repair suture through the shuttle loop; applying a pulling force to the first and second ends of the shuttle suture to pull the shuttle loop and the distal end of the repair suture through the pledget and a locking feature; and tensioning and locking the repair suture with the locking feature.

[0234] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget, a repair suture that may have a proximal end fixedly coupled to the pledget and a distal end coupled to a first needle, a locking feature that may be coupled to the pledget, and a shuttle suture that may have a first end coupled to the first needle and a second end coupled to a second needle. The repair suture may be configured to be inserted through the shuttle suture via the first needle and guided through the locking feature by the shuttle suture to retain the repair suture under tension when the repair suture is threaded through the locking feature.

[0235] According to an aspect of the present disclosure, a method of replacing a cardiovascular valve may include one or more of: inserting a first needle through tissue and a replacement valve, where the first needle is coupled to a distal end of a repair suture and a first end of a shuttle suture; inserting a second needle through the tissue and the replacement valve to position a pledget against an inner surface of the tissue, where the second needle is coupled to a second end of the shuttle suture; removing the first needle from the first end of the shuttle suture; inserting the distal end of the repair suture through the shuttle suture using the first needle; removing the first needle from the distal end of the repair suture and the second needle from the second end of the suture shuttle; applying a pulling force to the first end of the shuttle suture to pull the second end of the shuttle suture and the distal end of the repair suture through the pledget and a locking feature; and tensioning and locking the repair suture with the locking feature.

[0236] According to an aspect of the present disclosure, a knotless valve suture assembly may include a pledget, first and second needles, a repair suture that may have a proximal end fixedly coupled to the pledget and a distal end coupled to the first needle, and a shuttle suture that may extend through the pledget. The shuttle suture may be operably coupled with the first and second needles. The shuttle suture may include a shuttling feature. A locking feature may be coupled to the pledget. The locking feature may be configured to retain the repair suture under tension when the repair suture is threaded through the locking feature by the shuttle suture.

[0237] According to an aspect of the present disclosure, a first end of a shuttle suture may be coupled with a first needle, and a second end of the shuttle suture may be coupled with a second needle.

[0238] According to an aspect of the present disclosure, a shuttling feature may be a shuttle link defined proximate to a second end.

[0239] According to an aspect of the present disclosure, a shuttle suture may have a bifurcated section that defines a shuttle link.

[0240] According to an aspect of the present disclosure, a shuttling feature may be a shuttle receiving area.

[0241] According to an aspect of the present disclosure, a shuttle suture may be constructed of multiple strands. A repair suture is configured to be inserted between the multiple strands at a shuttle receiving area.

[0242] According to an aspect of the present disclosure, a first end and a second end of a shuttle suture may be coupled to a first needle.

[0243] According to an aspect of the present disclosure, a shuttle suture may be folded to form two limbs that extend through a pledget.

[0244] According to an aspect of the present disclosure, a shuttling feature may be a shuttle loop including a mid-section of a shuttle suture.

[0245] According to an aspect of the present disclosure, a mid-section of the shuttle suture forming a shuttle loop may be operably coupled with a second needle.

[0246] According to an aspect of the present disclosure, a shuttle loop may be operably coupled with a second needle via a connecting suture.

[0247] According to an aspect of the present disclosure, a method of replacing a cardiovascular valve may include one or more of: inserting a first needle through tissue and a replacement valve, wherein the first needle is coupled to a distal end of a repair suture and a first end of a shuttle suture; inserting a second needle through the tissue and the replacement valve, wherein the second needle is operably coupled to the shuttle suture; positioning a pledget against an inner surface of the tissue; removing the first needle and the second needle from the shuttle suture; inserting the distal end of the repair suture through a shuttling feature of the shuttle suture; applying a tension to at least the first end of the shuttle suture; drawings the shuttling feature and the distal end of the repair suture through the pledget and a locking feature coupled with the pledget; tensioning the repair suture with application of the tension; and locking the repair suture in a stitch configuration by engaging the locking feature.

[0248] According to an aspect of the present disclosure, a method may include removing the first needle from the repair suture.

[0249] According to an aspect of the present disclosure, a method may include a step of removing a first needle and a second needle from a shuttle suture, which may include cutting a connecting suture and removing the second needle and the connecting suture from the shuttle suture.

[0250] According to an aspect of the present disclosure, a method may include a step of applying a tension to at least a first end of a shuttle suture, which may include applying the tension to both the first end and a second end of the shuttle suture.

[0251] According to an aspect of the present disclosure, a method may include a step of inserting a distal end of a repair suture through a shuttling feature, which may include inserting the distal end through a folded shuttle loop formed from folding a shuttle suture.

[0252] According to an aspect of the present disclosure, a method may include a step of inserting a second needle through a tissue and a replacement valve, which may include drawing a mid-section of a shuttle suture through the tissue and the replacement valve with a second needle.

[0253] According to an aspect of the present disclosure, a method may include a step of inserting a second needle through a tissue and a replacement valve, which may include drawing a second end of a shuttle suture through the tissue and the replacement valve with a second needle.

[0254] According to an aspect of the present disclosure, a step of inserting a distal end of a repair suture through a shuttling feature may include inserting the distal end through a bifurcated shuttle link.

[0255] According to an aspect of the present disclosure, a shuttle suture may be formed of multiple strands of material. A method may include a step of inserting the distal end of a repair suture through a shuttling feature, which may include inserting the distal end between the multiple strands of material in a shuttle receiving area.

[0256] According to an aspect of the present disclosure, a locking feature may be a spliced section defined by a repair suture. A step of locking the repair suture in a stitch configuration may include drawing the repair suture through the spliced section and bunching stuffing in the repair suture proximate to the spliced section.

[0257] According to an aspect of the present disclosure, a locking feature may be a grommet with inner barbs. A step of locking a repair suture in a stitch configuration may include drawing the repair suture through the grommet and engaging the inner barbs.

[0258] According to an aspect of the present disclosure, a knotless suture assembly may include a repair suture including a first end and a second end and a shuttle suture having a first end and a second end. The first end may form a shuttling loop. The shuttle suture may be spliced into the repair suture at a first spliced segment proximate and a second spliced segment. A stuffing suture may be spliced into the repair suture between the first and second spliced segments. The shuttle suture may be configured to guide the first end of the repair suture through the first spliced segment which can, consequently, bunch the stuffing suture to lock the repair suture in a stitch configuration.

[0259] According to an aspect of the present disclosure, a first spliced segment may be proximate to a first end of a repair suture and a first end of a shuttle suture.

[0260] According to an aspect of the present disclosure, a second spliced segment may be proximate to a second end of a repair suture and a second end of a shuttle suture.

[0261] According to an aspect of the present disclosure, a second end of a shuttle suture may be encapsulated in a repair suture proximate to a second end of the repair suture.

[0262] According to an aspect of the present disclosure, a first spliced segment may be disposed proximate to a first transition section between multiple limbs of a repair suture on a first side of a mid-region of the repair suture. A second spliced segment may be disposed proximate to a second transition section between multiple limbs of the repair suture on a second side of the mid-region of the repair suture.

[0263] According to an aspect of the present disclosure, a mid-region may have a greater thickness than each of multiple limbs on each of first and second sides thereof.

[0264] According to an aspect of the present disclosure, first and second ends of a repair suture are tapered.

[0265] According to an aspect of the present disclosure, a method of manufacturing a suture assembly may include one or more of: cutting suture material to a predefined length; forming a repair suture with a mid-region, multiple limbs on a first side of the mid-region, and multiple limbs on a second side of the mid-region; splicing a shuttle suture into the repair suture at a first location; splicing a stuffing suture into the mid-region of the repair suture; and splicing the shuttle suture into the repair suture at a second location spaced from the first location; forming a first end of the repair suture with the multiple limbs on the first side; and forming a second end of the repair suture with the multiple limbs on the second side.

[0266] According to an aspect of the present disclosure, a method may include positioning a shuttle loop of the shuttle suture proximate to the first end of the repair suture and positioning a non-looped end of the shuttle suture proximate to the second end of the repair suture.
[0267] According to an aspect of the present disclosure, a step of splicing a shuttle suture into a repair suture at a first location may include splicing the shuttle suture into the repair suture proximate to a transition section between a mid-region and multiple limbs on a first side.
[0268] According to an aspect of the present disclosure, a step of splicing a shuttle suture into a repair suture at a second location may include splicing the shuttle suture into the repair suture proximate to a transition section between a mid-region and multiple limbs on a second side.
[0269] According to an aspect of the present disclosure, first and second locations may be on opposing sides of a mid-region.
[0270] According to an aspect of the present disclosure, a step of splicing a stuffing suture into a mid-region may include increasing a thickness of a repair suture at the mid-region.
[0271] According to an aspect of the present disclosure, a step of forming a repair suture may include splicing suture segments together to form multiple limbs on opposing sides of a mid-region.
[0272] According to an aspect of the present disclosure, a step of forming a repair suture may include bifurcating suture material to form multiple limbs.
[0273] According to an aspect of the present disclosure, a method of compressing tissue bodies may include one or more of: providing a suture assembly with a shuttle suture spliced into a repair suture to include a spliced segment; inserting a suture assembly through a first tissue body in a first direction; inserting the suture assembly through a second tissue body in a second opposing direction; engaging an end of the repair suture with a shuttle loop of the shuttle suture; guiding the end of the repair suture through the spliced segment; and applying a tension force to the end of the repair suture; compressing first and second tissue bodies together with the repair suture; and locking the repair suture under tension.
[0274] According to an aspect of the present disclosure, a method may include inserting the suture assembly through the second tissue body in the first direction and inserting the suture assembly through the first tissue body in the second opposing direction.
[0275] According to an aspect of the present disclosure, a first tissue body overlaps a second tissue body.
[0276] According to an aspect of the present disclosure, a first tissue body is laterally adjacent to a second tissue body.
[0277] According to an aspect of the present disclosure, a knotless suture assembly may include a repair suture including a mid-region between a first end and a second end and a shuttle suture including a mid-section between a first end and a second end. The first end of the shuttle suture may be spliced into the repair suture at a first spliced segment. The second end of the shuttle suture may include a shuttling feature spliced into the repair suture at a second spliced segment. A locking feature may include a third spliced segment formed between the mid-section and the mid-region. The locking feature may be configured to retain the repair suture under tension in response to a shuttling process. A first needle may be coupled to at least the first end of the repair suture. A second needle may be coupled to at least the second end of the repair suture.
[0278] According to an aspect of the present disclosure, a repair suture may extend beyond first and second ends of a shuttle suture to engage first and second needles, respectively.
[0279] According to an aspect of the present disclosure, at least one of first and second ends of a repair suture may be tapered to have a lesser thickness than a mid-region.
[0280] According to an aspect of the present disclosure, a pledget may be operably coupled with a locking feature.
[0281] According to an aspect of the present disclosure, a stuffing suture may be spliced into a repair suture at a spliced region proximate to a locking feature to increase a thickness of the repair

suture at the spliced region.
[0282] According to an aspect of the present disclosure, a spliced region may be disposed between a locking feature and a first spliced segment.
[0283] According to an aspect of the present disclosure, first and second ends of a shuttle suture may be encapsulated in a repair suture. First and second ends of the shuttle suture may be configured to be separated from a repair suture in response to removal of first and second According to an aspect of the present disclosure, a shuttling feature may be an end shuttle loop.
[0284] According to an aspect of the present disclosure, a method of manufacturing a knotless suture assembly may include one or more of: forming a repair suture with a mid-region between a first end and a second end; providing a shuttle suture with a first end and a second end with a shuttling feature; splicing the first end of the shuttle suture into the repair suture at a first location proximate to the first end of the repair suture; forming a first spliced segment at the first location between the shuttle suture and the repair suture; splicing the shuttle suture into the repair suture at a second location spaced from the first location; forming a second spliced segment between the shuttle suture and the repair suture at the second location; forming a locking feature with the second spliced segment for retaining the repair suture under tension in response to a shuttling process; and coupling the first end of the repair suture with a first needle.
[0285] According to an aspect of the present disclosure, a method may include splicing a second end of a shuttle suture into a repair suture proximate to a second end of the repair suture and forming a third spliced segment between the shuttle suture and the repair suture at a third location.
[0286] According to an aspect of the present disclosure, a method may include coupling a second end of a repair suture with a second needle.
[0287] According to an aspect of the present disclosure, a method may include splicing a stuffing suture into a mid-region of a repair suture on an opposing side of a locking feature relative to a shuttling feature.
[0288] According to an aspect of the present disclosure, a method may include forming a first end of the repair suture with multiple limbs on a first side of a mid-region and forming a second end of the repair suture with multiple limbs on a second side of the mid-region.
[0289] According to an aspect of the present disclosure, a step of splicing a shuttle suture into a repair suture at the second location may include splicing a mid-section of the shuttle suture into a mid-region of the repair suture.
[0290] According to an aspect of the present disclosure, a method may include removing a first needle and separating a first end of a shuttle suture from a first end of a repair suture.
[0291] According to an aspect of the present disclosure, a knotless suture assembly may include a repair suture including a first end and a second end and a shuttle suture having a first end and a second end. The second end of the shuttle suture may form a shuttling feature. The shuttle suture may be spliced into the repair suture at a first spliced segment proximate to the first end of the repair suture and at a second spliced segment. The second spliced segment forms at least a portion of a locking feature. A stuffing suture may be spliced into the repair suture between the first and second spliced segments. The shuttling feature may be configured to guide the first end of the repair suture through the locking feature in response to a pulling force being applied to the first end of the shuttle suture which may, consequently, bunch the stuffing suture to lock the repair suture in a stitch configuration. At least one needle may be coupled with the repair suture.
[0292] According to an aspect of the present disclosure, a second end of a shuttle suture with a shuttling feature may be spliced into a repair suture proximate to a second end of the repair suture.
[0293] According to an aspect of the present disclosure, at least one needle may include a first needle coupled to a first end of a repair suture and a second needle coupled to a second end of the repair suture.
[0294] According to an aspect of the present disclosure, a pledget may be operably coupled with a repair suture proximate to a locking feature.

[0295] According to an aspect of the present disclosure, first and second ends of a repair suture may be tapered.
[0296] According to an aspect of the present disclosure, a method of attaching a cardiovascular replacement valve to tissue may include one or more of: inserting a first needle through tissue and a replacement valve; carrying a first end of a repair suture and a first end of a shuttle suture through the tissue and the replacement valve with the first needle; inserting a second needle through the tissue and the replacement valve; carrying a second end of the shuttle suture through the tissue and the replacement valve with the second needle; positioning a pledget against an inner surface of the tissue; removing the first and second needles; inserting the first end of the repair suture through a shuttling feature at the second end of the shuttle suture; applying a pulling force to the first end of the shuttle suture; pulling the shuttling feature and the first end of the repair suture through a locking feature; and locking the repair suture under tension with the locking feature.
[0297] According to an aspect of the present disclosure, a method may include providing a shuttle suture having a shuttling feature configured as a bifurcated segment forming a shuttle link.
[0298] According to an aspect of the present disclosure, a step of inserting a first end of a repair suture through a shuttling feature may include inserting the first end of the repair suture through a shuttle link.
[0299] According to an aspect of the present disclosure, a step of inserting a first end of a repair suture through a shuttling feature may include inserting the first end of the repair suture between strands of a shuttle suture at a shuttle receiving area.
[0300] According to an aspect of the present disclosure, a method may include one or more of: pulling a first end of a repair suture across a replacement valve, through the replacement valve, through tissue and to a locking feature; pulling the first end of the repair suture from the locking feature, through the tissue, and through the replacement valve; forming a stitch configuration with the repair suture; and sealing the replacement valve to the tissue with the repair suture in the stitch configuration.
[0301] According to an aspect of the present disclosure, a method may include tensioning a repair suture and re-tensioning the repair suture, wherein a locking feature may be configured to lock the repair suture under tension after each re-tensioning of the repair suture.
[0302] According to an aspect of the present disclosure, a method may include carrying a second end of a repair suture through tissue and a replacement valve with a second needle.
[0303] According to an aspect of the present disclosure, a locking feature may be a spliced section defined by a repair suture. A step of locking the repair suture under tension may include one or more of: drawing the repair suture through the spliced section; forming a stitch configuration with the repair suture; and bunching stuffing in the repair suture proximate the spliced section.
[0304] According to an aspect of the present disclosure, a locking feature may be a grommet with inner barbs. A step of locking the repair suture in under tension may include one or more of: drawing a repair suture through the grommet; forming a stitch configuration with the repair suture; and engaging the inner barbs.
[0305] According to an aspect of the present disclosure, a method of replacing a cardiovascular valve may include one or more of: inserting a first needle through tissue and a replacement valve; carrying an end of a repair suture, a first end of a shuttle suture, and a second end of the shuttle suture through the tissue and the replacement valve with the first needle; inserting a second needle through the tissue and the replacement valve; carrying a mid-section of the shuttle suture through the tissue and the replacement valve with the second needle; positioning a pledget against an inner surface of the tissue; removing the first needle and the second needle; inserting the end of the repair suture through a shuttling feature of the shuttle suture; applying a tension to at least the first end of the shuttle suture; drawing the shuttling feature and the end of the repair suture through a locking feature operably coupled with the pledget; tensioning the repair suture; and locking the repair suture under tension in a stitch configuration in response to an engagement between the repair

suture and the locking feature.
[0306] According to an aspect of the present disclosure, a step of removing a first needle and a second needle may include one or more of: removing the first needle from a shuttle suture and a repair suture; cutting a connecting suture; and removing the second needle and the connecting suture from the shuttle suture.
[0307] According to an aspect of the present disclosure, a step of applying tension to at least a first end of a shuttle suture may include applying the tension to both the first end and a second end of the shuttle suture and pulling the mid-section of the shuttle suture through a locking feature.
[0308] According to an aspect of the present disclosure, a step of inserting an end of a repair suture through a shuttling feature may include inserting the end through a folded shuttle loop formed from folding a shuttle suture at a mid-section.
[0309] According to an aspect of the present disclosure, a locking feature may be a spliced section defined by a repair suture. A step of locking the repair suture in a stitch configuration may include drawing the repair suture through the spliced section and bunching stuffing in the repair suture proximate to the spliced section.
[0310] According to an aspect of the present disclosure, a locking feature may be a grommet with inner barbs. A step of locking the repair suture in a stitch configuration may include drawing the repair suture through the grommet and engaging the inner barbs.
[0311] According to an aspect of the present disclosure, a method of compressing cardiovascular tissue bodies may include one or more of: providing a suture assembly with a repair suture and a shuttle suture; inserting an end of the suture assembly through a first tissue body in a first direction; inserting the end of the suture assembly through a second tissue body in a second opposing direction; engaging an end of the repair suture with a shuttling feature of the shuttle suture; a spliced segment between the shuttle suture and the repair suture; applying a tension force to the end of the repair suture; compressing first and second tissue bodies together with the repair suture; and locking the repair suture under tension.
[0312] According to an aspect of the present disclosure, a method may include inserting a suture assembly through a second tissue body in a first direction and inserting the suture assembly through a first tissue body in a second opposing direction.
[0313] According to an aspect of the present disclosure, a step of inserting an end a suture assembly through a second tissue body may include inserting the end of the suture assembly through the second tissue body that overlaps a first tissue body.
[0314] According to an aspect of the present disclosure, a step of inserting an end a suture assembly through a second tissue body may include inserting the end of the suture assembly through the second tissue body that is laterally adjacent to a first tissue body.
[0315] According to an aspect of the present disclosure, a first tissue body overlaps a second tissue body.
[0316] According to an aspect of the present disclosure, a first tissue body is laterally adjacent to a second tissue body.
[0317] According to an aspect of the present disclosure, a method may include providing a suture assembly with a shuttle suture spliced into a repair suture to include at least one spliced segment.
[0318] According to an aspect of the present disclosure, a step of providing a suture assembly may include providing the suture assembly with a shuttle suture spliced into a repair suture at multiple locations.
[0319] It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present device. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.
[0320] It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present device,

and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

[0321] The above description is considered that of the illustrated embodiments only. Modifications of the device will occur to those skilled in the art and to those who make or use the device. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the device, which is defined by the following claims as interpreted according to the principles of patent law, including the Doctrine of Equivalents.

Claims

1. A knotless suture assembly, comprising: a repair suture including a mid-region between a first end and a second end; a shuttle suture including a mid-section between a first end and a second end, wherein the first end of the shuttle suture is spliced into the repair suture at a first spliced segment, and wherein the second end of the shuttle suture includes a shuttling feature spliced into the repair suture at a second spliced segment; a locking feature including a third spliced segment formed between the mid-section and the mid-region, the locking feature configured to retain the repair suture under tension in response to a shuttling process; a first needle coupled to at least the first end of the repair suture; and a second needle coupled to at least the second end of the repair suture.

2. The knotless suture assembly of claim 1, wherein the repair suture extends beyond the first and second ends of the shuttle suture to engage the first and second needles, respectively.

3. The knotless suture assembly of claim 1, wherein at least one of the first and second ends of the repair suture are tapered to have a lesser thickness than the mid-region.

4. The knotless suture assembly of claim 1, further comprising: a pledget operably coupled with the locking feature.

5. The knotless suture assembly of claim 1, further comprising: a stuffing suture spliced into the repair suture at a spliced region proximate to the locking feature to increase a thickness of the repair suture at the spliced region.

6. The knotless suture assembly of claim 5, wherein the spliced region is disposed between the locking feature and the first spliced segment.

7. The knotless suture assembly of claim 1, wherein the first and second ends of the shuttle suture are encapsulated in the repair suture, and wherein the first and second ends of the shuttle suture are configured to be separated from the repair suture in response to removal of the first and second needles.

8. The knotless suture assembly of claim 1, wherein the shuttling feature is an end shuttle loop.

9. A method of manufacturing a knotless suture assembly, comprising: forming a repair suture with a mid-region between a first end and a second end; providing a shuttle suture with a first end and a second end with a shuttling feature; splicing the first end of the shuttle suture into the repair suture at a first location proximate to the first end of the repair suture; forming a first spliced segment at the first location between the shuttle suture and the repair suture; splicing the shuttle suture into the repair suture at a second location spaced from the first location; forming a second spliced segment between the shuttle suture and the repair suture at the second location; forming a locking feature with the second spliced segment for retaining the repair suture under tension in response to a shuttling process; and coupling the first end of the repair suture with a first needle.

10. The method of claim 9, further comprising: splicing the second end of the shuttle suture into the repair suture proximate to the second end of the repair suture; and forming a third spliced segment between the shuttle suture and the repair suture at a third location.

11. The method of claim 10, further comprising: coupling the second end of the repair suture with a second needle.

12. The method of claim 9, further comprising: splicing a stuffing suture into the mid-region of the repair suture on an opposing side of the locking feature relative to the shuttling feature.
13. The method of claim 9, further comprising: forming the first end of the repair suture with multiple limbs on a first side of the mid-region; and forming the second end of the repair suture with multiple limbs on a second side of the mid-region.
14. The method of claim 9, wherein the step of splicing the shuttle suture into the repair suture at the second location includes splicing a mid-section of the shuttle suture into the mid-region of the repair suture.
15. The method of claim 9, further comprising: removing the first needle; and separating the first end of the shuttle suture from the first end of the repair suture.
16. A knotless suture assembly, comprising: a repair suture including a first end and a second end; a shuttle suture having a first end and a second end, the second end of the shuttle suture forming a shuttling feature, wherein the shuttle suture is spliced into the repair suture at a first spliced segment proximate the first end of the repair suture and at a second spliced segment, wherein the second spliced segment forms at least a portion of a locking feature; a stuffing suture spliced into the repair suture between the first and second spliced segments, wherein the shuttling feature is configured to guide the first end of the repair suture through the locking feature in response to a pulling force being applied to the first end of the shuttle suture which, consequently, bunches the stuffing suture to lock the repair suture in a stitch configuration; and at least one needle coupled with the repair suture.
17. The knotless suture assembly of claim 16, wherein the second end of the shuttle suture with the shuttling feature is spliced into the repair suture proximate to the second end of the repair suture.
18. The knotless suture assembly of claim 16, wherein the at least one needle includes a first needle coupled to the first end of the repair suture and a second needle coupled to the second end of the repair suture.
19. The knotless suture assembly of claim 16, further comprising: a pledget operably coupled with the repair suture proximate to the locking feature.
20. The knotless suture assembly of claim 16, wherein the first and second ends of the repair suture are tapered.